TEACHER'S CLASSROOM SCRIPT

Bayesian Inference

A Continuous Professor-Led Teaching Narrative

One connected lecture from first principles to hierarchical models

Complete theory, 35 questions, and worked solutions woven into a continuous teaching narrative Designed for live instruction: the professor's spoken narrative accompanies definitions, derivations, examples, practice questions and complete solutions.

Prepared from the uploaded *Bayesian Inference, up to Week 13* lecture notes and the eight-unit Bayesian Inference syllabus supplied with the course materials.

A self-contained teaching script designed to be read aloud, adapted into a lecture, or used as a written substitute for spoken explanation. Mathematical conventions and assumptions are stated explicitly.

How to use this guide

This manual is written as one continuous classroom transcript rather than as a collection of disconnected explanations or terse revision notes. The intention is that a teacher can read the explanation aloud, use the displayed mathematics as board-work, pause at the indicated teaching cues, and move naturally from intuition to formal derivation. The script is deliberately explicit: it states what each symbol means, why a step is valid, what the result tells us, and which common misunderstanding should be avoided.

The eight theory chapters follow the supplied syllabus in order. Every section and subsection is accompanied by a spoken explanation. The seven question types then provide a teacher's walkthrough for each of the 35 questions. Questions labelled as practice are first introduced with a teaching plan; their complete worked solutions are collected in the final solutions chapter, where the mathematical steps are accompanied by a spoken explanation. This arrangement lets the teacher choose between presenting a question live, assigning it for independent work, or reading the solution as a guided demonstration.

The manual is designed to be self-contained. It does not require the teacher or student to consult another text to fill in missing definitions or algebraic steps. Where the original lecture notes only posed a derivation or gave a brief mention, the full-syllabus material already incorporated in the source notes is retained and explained in teaching language.

What was used as the source, and what was expanded

The uploaded lecture PDF provides the course's framing and worked foundations: classical-versus-Bayesian inference, Bayes' theorem, the diagnostic-test and three-prisoners examples, likelihood principle, the three steps of Bayesian data analysis, exchangeability and de Finetti's theorem, Beta–Bernoulli updating and prediction, Normal–Normal updating, prior and predictive distributions, exponential-family conjugacy, improper and Jeffreys priors, nuisance-parameter marginalization, scaled-inverse-chi-square distributions, MAP and credible intervals, posterior odds and Bayes factors, Bayesian linear regression, ridge/LASSO priors, spike-and-slab selection, Gibbs sampling, and random-effects motivation. Several later pages are handwritten scans, so their content is reflected in the explanations rather than relying only on searchable text.

The syllabus explicitly names some models and methods for which the lecture PDF gives only a brief mention, a prompt to derive them, or no complete worked treatment. To make the guide self-sufficient, the Gamma–Exponential, Gamma–Poisson, Dirichlet–Multinomial, full Normal–Inverse-Gamma/Normal–Gamma, detailed Metropolis–Hastings procedure and diagnostics, empirical Bayes, and profile-likelihood treatments are supplied here as syllabus-completion explanations. These are clearly standard derivations consistent with the syllabus, not claims that every derivation appeared verbatim in the uploaded slides.

Exam check

A good exam solution normally follows this order: (i) state the sampling model, (ii) write the prior, (iii) write the likelihood, (iv) multiply prior and likelihood, (v) identify the posterior kernel, (vi) state the normalized posterior distribution and its parameters, and (vii) answer the quantity actually requested. Always name the posterior before calculating.

The story that connects the whole course

Teacher's spoken explanation

Before we write a single formula, I want you to imagine that we have been asked to advise a large group of hospitals. The hospitals are collecting information about patients: whether a patient recovers, how long a patient waits, how many events occur in a given period, measurements such as blood pressure, and the characteristics that may help explain different outcomes. The administrators do not merely want a table of numbers. They want to learn from incomplete and variable evidence, quantify uncertainty, compare explanations, predict future outcomes, and understand why hospitals differ from one another.

That is the story we will follow throughout this course. At first, the question is very modest: what should we believe about an unknown probability after observing a few outcomes? That takes us to Bayes' theorem. As the data become richer, we discover that different kinds of observations lead to different likelihoods, and that carefully chosen priors can make updating exact. Once we have a posterior distribution, we must decide how to summarize it, which leads to Bayesian estimation. When the question changes from estimating a quantity to comparing claims, we arrive at posterior odds and Bayes factors. We then pause and ask where priors come from and how sensitive our conclusions are to them.

For simple models, algebra may give us the posterior directly. For complicated models, it may not. That practical difficulty leads naturally to Markov chain Monte Carlo. Once we can describe uncertainty computationally, we can model outcomes using several explanatory variables, regularize unstable estimates, and select variables. Finally, patients are not collected from one abstract population: they are treated in wards and hospitals. Their outcomes share common environments. That brings us to hierarchical models, partial pooling and empirical Bayes.

Notice the direction of travel. We are not studying eight unrelated chapters. We are taking one scientific investigation and asking progressively more demanding questions of it. Each new method appears because the previous question has become too limited. Keep asking, as we move forward: what problem are we trying to solve now, and what idea from the previous lecture makes the next step possible?

The running story is an illustration, not an extra model assumption

The hospital research setting is a pedagogical thread used to connect the syllabus. A binary recovery indicator motivates Bernoulli and Binomial models; event counts motivate Poisson models; waiting times motivate Exponential models; categories motivate the Multinomial model; measurements motivate Normal models; multiple predictors motivate regression; and groups of hospitals motivate hierarchical models. These are different data structures within one investigation, not a claim that one probability distribution describes every variable. The numerical examples and formal assumptions in each topic remain the ones stated in that topic.

Coverage audit of the original lecture PDF

The uploaded *Bayesian_upto_W13.pdf* contains 73 PDF pages, including handwritten pages that are not fully represented in its searchable text. The audit below therefore uses both the extracted text and visual inspection of the handwritten pages. "Covered" means the idea or derivation is present in the uploaded notes; "partial" means it is only mentioned, posed as an exercise, or not developed far enough to support a new examination problem; "missing" means no substantive treatment was found.

Unit	Core syllabus content	Original-note status	Evidence and gap
1	Bayes' theorem, motivation, prior, likelihood, posterior and examples	Strong	PDF pp. 1–5 and 6–12: classical/Bayesian contrast, disease test, three-prisoners example, posterior updating, exchangeability and Beta–Bernoulli foundations.
2	Exponential families and all listed conjugate models	Partial	PDF pp. 12–30 introduce exponential-family form, Beta–Bernoulli and Normal conjugacy; later notes include Normal–Inverse-chi-square / Normal–Inverse-Gamma and a Poisson-regression likelihood. Gamma–Exponential and Gamma–Poisson updates are not worked through systematically; Dirichlet–Multinomial is absent; the multi-parameter exponential-family discussion is brief.
3	Posterior point estimates, MAP and credible intervals	Partial	PDF pp. 10–12 give Beta posterior summaries; handwritten PDF p. 42 names MAP and credible intervals, and PDF p. 50 applies them to regression. A general loss-based account and systematic interval construction are not developed.
4	Bayesian tests, posterior odds and Bayes factors	Strong	Handwritten PDF pp. 42–48 develop posterior odds, simple and composite hypotheses, marginal likelihoods and Bayes-factor comparisons.
5	Conjugate, non-informative and Jeffreys priors	Strong	PDF pp. 13–30 and 27–30 discuss conjugacy, improper/flat priors, reparameterization and Jeffreys' prior with Fisher-information examples.

6	MCMC, Gibbs, Metropolis–Hastings and graphical convergence diagnostics	Major gaps	MCMC and Metropolis–Hastings are mentioned; Gibbs updates appear in the Normal and regression derivations (including handwritten PDF pp. 37–38, 56–57 and 66, 69–71). A complete MH acceptance-ratio treatment and graphical convergence diagnostics are not present.
7	Bayesian linear models, Normal–Inverse-Gamma, ridge, LASSO, spike-and-slab and variable selection	Strong, with handwritten derivations	Handwritten PDF pp. 50–62 develop Bayesian linear regression and Normal–Inverse-Gamma updating; pp. 64–66 connect ridge/LASSO to priors and Gibbs updates; pp. 69–71 introduce spike-and-slab variable selection. The handwriting is compressed, so this guide rewrites the derivations step by step.
8	Hierarchical Bayes, Empirical Bayes and profile likelihood	Major gaps	Handwritten PDF p. 72 introduces a random-effects model, but a full hierarchical-Bayes treatment, empirical-Bayes estimation and profile likelihood are not substantively covered.

On a checklist of 30 major topic clusters, with partial credit where a topic is only mentioned or incompletely developed, the original lecture PDF covers approximately 73%. This is a topic-level estimate rather than an exact measurement. The main examination risks are the incomplete conjugate-model family in Unit 2, formal posterior summaries in Unit 3, MCMC algorithms and diagnostics in Unit 6, and Empirical Bayes/profile likelihood in Unit 8.

How this full-syllabus guide handles the gaps

The next chapters retain and explain the lecture-note material, including handwritten derivations, while adding complete syllabus-level treatments for the underdeveloped or missing topics. In particular, the guide supplies worked Gamma–Exponential, Gamma–Poisson and Dirichlet–Multinomial updates; a full Normal–Gamma precision parameterization; posterior decision summaries and credible intervals; the Metropolis–Hastings algorithm and graphical diagnostics; and hierarchical Bayes, Empirical Bayes and profile likelihood. These are syllabus-completion explanations, not claims that every derivation appeared in the uploaded lecture PDF.

Syllabus map and learning checklist

The Bayesian Inference syllabus supplied with the course materials lists eight units. The following checklist translates them into the concrete skills expected in a written examination.

Unit	Syllabus area	You should be able to do this without notes
1	Background and motivation	Explain Bayes' theorem; identify prior, likelihood, evidence and posterior; compute posterior probabilities; explain why Bayesian inference conditions on observed data.
2	Exponential families and conjugacy	Recognize exponential-family form; identify sufficient statistics and natural parameters; derive conjugate updates for Beta–Bernoulli/Binomial, Gamma–Exponential, Gamma–Poisson, Normal–Gamma/Inverse-Gamma and Dirichlet–Multinomial models.
3	Bayesian estimation	Compute posterior means, medians, modes/MAP estimates and credible intervals; distinguish MAP from MLE; calculate posterior predictive distributions.
4	Bayesian hypothesis testing	Compute posterior odds, Bayes factors and marginal likelihoods; distinguish simple from composite hypotheses; explain how priors on parameters enter a composite-model Bayes factor.
5	Choice of priors	Compare conjugate, flat, diffuse, non-informative and Jeffreys priors; derive Jeffreys' prior from Fisher information; check whether a posterior is proper when the prior is improper.
6	Posterior simulation	Explain Markov chains, Gibbs sampling and Metropolis–Hastings; calculate an MH acceptance probability; interpret trace plots, autocorrelation plots, density overlays, running means and convergence summaries.
7	Bayesian linear models	Derive the Normal posterior for regression coefficients; connect Gaussian priors to ridge and Laplace priors to LASSO; explain Normal–Inverse-Gamma regression, spike-and-slab priors and posterior inclusion probabilities.
8	Hierarchical models	Factor a hierarchical joint distribution; explain complete/no/partial pooling; derive normal-normal shrinkage; distinguish hierarchical Bayes from empirical Bayes; define and construct a profile likelihood.

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Chapter 1

Bayesian foundations and motivation

Teacher's spoken explanation

Let us begin where every scientific investigation begins: with uncertainty. Our hospital team has seen a small number of patients, but it must make a statement about a much larger population. A sample proportion is easy to calculate; the harder question is how uncertain we should remain about the underlying recovery probability. Classical inference gives us one language for this task. Bayesian inference gives us another: it treats uncertainty about the unknown parameter as a distribution and updates that distribution when evidence arrives. In this first chapter we will build the grammar of that update. Every later chapter will use the same grammar, even when the likelihood and parameter become more complicated.

Teacher's spoken explanation

Class, today we are going to build Bayesian inference from the ground up. We will ask what is known, what is unknown, what information the data provide, and how uncertainty changes after observing the data. Keep the distinction between a model and the reality it represents in mind throughout the lecture.

1.1 What is Bayesian inference?

Teacher's spoken explanation

Let us begin with the central question of the course: what do we mean when we say that we are learning about an unknown quantity from data? In Bayesian inference, we represent uncertainty about the parameter by a probability distribution. Before seeing the current data, that distribution is the prior. After observing the data, we update it to obtain the posterior. The point is not that the parameter itself is random in every scientific interpretation; rather, probability is being used to represent uncertainty about its possible values.
As we proceed, keep the three ingredients separate. The prior describes what was plausible before this observation. The likelihood describes how compatible the observed data are with each possible parameter value. The posterior combines both. Whenever we solve a problem, we will identify these three pieces before doing algebra.

Statistical inference concerns unknown features of a population or data-generating mechanism. Let θ denote an unknown parameter and let $Y = (Y_1, \ldots, Y_n)$ denote observed data. A statistical model specifies the probability distribution of the data conditional on the parameter, written $p(y \mid \theta)$ or $f(y \mid \theta)$.

The Bayesian approach treats uncertainty about θ probabilistically. Before observing the current data, uncertainty is represented by a **prior distribution** $p(\theta)$. After observing y, uncertainty is updated

to the **posterior distribution** $p(\theta \mid y)$.

Bayes' theorem: the central updating equation

$$p(\theta \mid y) = \frac{p(\theta)p(y \mid \theta)}{p(y)}, \qquad p(y) = \int p(\theta)p(y \mid \theta)\,\mathrm{d}\theta.$$

For a discrete parameter, replace the integral by a sum. The denominator $p(y)$ is the marginal likelihood or evidence. It is the normalizing constant that makes the posterior integrate or sum to one.

The four ingredients must not be confused:

- **Prior** $p(\theta)$: uncertainty about θ before the current data are observed.
- **Likelihood** $p(y \mid \theta)$: the probability or density of the observed data viewed as a function of θ. It is not a probability distribution over θ.
- **Evidence** $p(y)$: the prior-predictive probability/density of the observed data; it normalizes the posterior and is central to model comparison.
- **Posterior** $p(\theta \mid y)$: updated uncertainty about θ after observing y.

1.1.1 Kernel form: posterior proportional to prior times likelihood

Teacher's spoken explanation

Now let us turn Bayes' theorem into the working formula we use most often. The marginal probability of the observed data is the normalizing constant. It does not depend on the unknown parameter, so when our task is to identify the shape of the posterior as a function of that parameter, we can temporarily leave this constant aside. That gives the posterior-kernel rule: posterior is proportional to prior multiplied by likelihood.
I want you to notice the word proportional. It does not mean that we have finished finding a probability density. We have found its shape, but we must still identify the distribution and its normalizing constant, or calculate that constant if no familiar family appears.

For most derivations the evidence does not depend on θ. Therefore, when deriving the posterior as a function of θ, we can write

$$p(\theta \mid y) \propto p(\theta)p(y \mid \theta).$$

The proportionality sign means that a constant independent of θ has been omitted. It does not mean that the posterior is allowed to be unnormalized. A valid posterior density must integrate to one; the prior may be an improper density kernel if the resulting normalizing integral is finite.

1.2 A numerical Bayes' theorem example: diagnostic testing

Teacher's spoken explanation

This example is useful because it exposes a very common error. A test can be highly sensitive and still have a modest positive predictive value when the disease is rare. We must not confuse the probability of a positive result among diseased people with the probability of disease among people who tested positive. Those are reversed conditional probabilities.
We will first calculate the overall chance of a positive test using the law of total probability. Only then will we divide the joint probability of disease and a positive result by the probability of a positive result. Keep the denominator visible; it accounts for positive results in both diseased and non-diseased people.

Suppose a disease has prevalence $\mathbb{P}(D) = 0.02$. A test has sensitivity $\mathbb{P}(T^+ \mid D) = 0.95$ and false-positive probability $\mathbb{P}(T^+ \mid D^c) = 0.10$. The question is the probability that a person has the disease after a positive result.

First compute the marginal probability of a positive result by the law of total probability:

$$\mathbb{P}(T^+) = \mathbb{P}(D)\mathbb{P}(T^+ \mid D) + \mathbb{P}(D^c)\mathbb{P}(T^+ \mid D^c) = 0.02(0.95) + 0.98(0.10) = 0.117.$$

Then apply Bayes' theorem:

$$\mathbb{P}(D \mid T^+) = \frac{0.02(0.95)}{0.117} = \frac{0.019}{0.117} \approx 0.1624.$$

A positive result gives a posterior disease probability of about 16.24%, not 95%. The value 95% is the probability of a positive test conditional on disease, not the probability of disease conditional on a positive test. The difference is caused by the low prior prevalence and the false positives among the much larger disease-free population.

Diagnostic-test interpretation

Sensitivity is $\mathbb{P}(T^+ \mid D)$. Specificity is $\mathbb{P}(T^- \mid D^c)$. The false-positive rate is $1 -$ specificity $= \mathbb{P}(T^+ \mid D^c)$. The positive predictive value is $\mathbb{P}(D \mid T^+)$. These are different conditional probabilities and must not be interchanged.

1.3 Prior odds, posterior odds and likelihood ratios

Teacher's spoken explanation

Here we express Bayesian updating in odds form. Odds compare two alternatives directly. The data change those odds by a multiplicative factor called the Bayes factor, which is a ratio of likelihoods or marginal likelihoods depending on whether the hypotheses are simple or composite. The teaching point is that prior odds and the Bayes factor have different jobs. Prior odds describe our starting balance between the hypotheses. The Bayes factor describes how strongly the observed data favour one hypothesis over the other. Multiplying them gives posterior odds.

For two possible parameter values or two hypotheses H_1 and H_0, Bayes' theorem gives

$$\frac{\mathbb{P}(H_1 \mid y)}{\mathbb{P}(H_0 \mid y)} = \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \times \frac{p(y \mid H_1)}{p(y \mid H_0)}.$$

The first ratio is the **prior odds**; the ratio of data probabilities is the **Bayes factor** BF_{10}; the left-hand side is the **posterior odds**.

For two simple hypotheses, each hypothesis specifies the data distribution completely, so

$$\mathrm{BF}_{10} = \frac{p(y \mid H_1)}{p(y \mid H_0)}.$$

If $\mathrm{BF}_{10} > 1$, the observed data are more probable under H_1 than under H_0. If $\mathrm{BF}_{10} < 1$, they are more probable under H_0. The Bayes factor changes prior odds into posterior odds; the Bayes factor itself does not contain the prior odds on the hypotheses.

1.4 The three-prisoners example and the importance of the data-collection rule

Teacher's spoken explanation

This problem teaches us that probability statements depend on how information is generated. We cannot calculate the conditional probability merely from the sentence spoken by the guard. We must also model the guard's rule for choosing whom to name.
Let us list the possible pardoned prisoners, then ask what the guard would say in each case. This makes the conditional probabilities explicit. The example is a warning against treating information as if it had arrived through a neutral channel when the selection mechanism is actually part of the experiment.

Three prisoners A, B, C have equal prior probabilities of being pardoned. Prisoner A asks the guard to name one of B, C who will be executed. If both B and C will be executed, assume the guard chooses between them with equal probability. If the guard says "B will be executed", then

$$\begin{aligned}\mathbb{P}(\text{guard says B} \mid A \text{ pardoned}) &= \tfrac{1}{2},\\ \mathbb{P}(\text{guard says B} \mid B \text{ pardoned}) &= 0,\\ \mathbb{P}(\text{guard says B} \mid C \text{ pardoned}) &= 1.\end{aligned}$$

Thus

$$\mathbb{P}(A \text{ pardoned} \mid \text{guard says B}) = \frac{(1/3)(1/2)}{(1/3)(1/2) + (1/3)(0) + (1/3)(1)} = \frac{1}{3}.$$

The probability does not become 1/2. The guard's selection mechanism is part of the probability model. A conditional-probability problem is incomplete until the data-collection or reporting mechanism is specified.

1.5 Bayesian data analysis: the three steps

Teacher's spoken explanation

A complete Bayesian analysis is more than writing Bayes' theorem once. First we specify a probability model for every observable and unknown quantity. Second we condition on the observed data to obtain the posterior. Third we examine whether the model can reproduce the features of the data that matter scientifically.
The third step is essential. A posterior can be calculated exactly under a model that is nevertheless a poor description of reality. We therefore separate mathematical updating from model checking.

The lecture notes organize Bayesian analysis into three steps:

1. **Set up a full probability model.** Specify a joint distribution for observable quantities and unknown quantities, including the data-collection mechanism. For a coin with unknown success probability θ, a basic model is

   $$\theta \sim p(\theta), \qquad Y_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

   so that

   $$p(\theta, y_1, \ldots, y_n) = p(\theta) \prod_{i=1}^{n} p(y_i \mid \theta).$$

   The observed variables are $y_1, \ldots, y_n$; θ is unknown.

2. **Condition on the observed data.** Use Bayes' theorem to obtain $p(\theta \mid y)$ and any other conditional distributions of interest.

3. **Evaluate model fit.** Ask whether replicated data from the fitted model resemble the observed data in scientifically important ways. Posterior predictive checks compare observed summaries or plots with the same summaries or plots computed from data simulated under the posterior predictive distribution.

1.6 Conditional independence, exchangeability and de Finetti's theorem

Teacher's spoken explanation

The next idea explains why Bayesian models often begin by saying that observations are independent conditional on a parameter. Conditional independence is not the same as unconditional independence. Once we integrate out a shared unknown parameter, observations can become dependent because they share uncertainty about that parameter.
Exchangeability gives another way to describe a sequence: its joint distribution is unchanged by reordering the observations. For infinitely exchangeable binary sequences, de Finetti's theorem connects this symmetry to a mixture of iid Bernoulli models. This is the conceptual bridge from an exchangeable sequence to the familiar prior-plus-conditional-sampling model.

1.6.1 Conditional independence given a parameter

Teacher's spoken explanation

Imagine that the unknown success probability is fixed at a particular value of theta. Conditional on that value, the tosses are independent Bernoulli observations. But if theta itself is uncertain, observing the first few tosses changes what we believe about theta, and therefore changes our prediction for the next toss. The dependence appears after averaging over theta.

In a common Bayesian sampling model, observations are independent once the parameter is known:

$$p(y_{n+1} \mid \theta, y_1, \ldots, y_n) = p(y_{n+1} \mid \theta).$$

This does not normally imply unconditional independence. After integrating out θ,

$$p(y_1, \ldots, y_n) = \int \left\{ \prod_{i=1}^{n} p(y_i \mid \theta) \right\} p(\theta)\, \mathrm{d}\theta,$$

and the shared uncertain parameter can induce dependence among the observations. For example, if several coin tosses share the same unknown success probability, observing heads increases the posterior belief that θ is large and therefore changes the predictive probability of another head.

1.6.2 Exchangeability

Teacher's spoken explanation

Exchangeability means that the order in which we write the observations does not change their joint probability. It is a statement about symmetry of the joint distribution, not a claim that the observations are independent. We should be able to explain both directions carefully: iid observations are exchangeable, but exchangeability alone does not generally imply unconditional independence.

A finite sequence $Y_1, \ldots, Y_n$ is exchangeable if its joint distribution is unchanged by any permutation of the indices:

$$p(y_1, \ldots, y_n) = p(y_{\pi(1)}, \ldots, y_{\pi(n)})$$

for every permutation π. The labels or order of the observations do not matter; only the multiset of values matters. Every iid sequence is exchangeable. The converse is not generally true without additional assumptions.

A simple finite example is drawing two different numbered objects without replacement from a set containing one red and one blue object. The two draws are exchangeable, because the probabilities of red-then-blue and blue-then-red are equal, but the draws are not independent: after a red first draw, the second draw must be blue.

1.6.3 De Finetti's representation theorem

Teacher's spoken explanation

For an infinitely extendable exchangeable sequence of binary outcomes, de Finetti's theorem says that its joint law can be represented as a mixture of iid Bernoulli laws. The mixing variable is theta, and its distribution plays the role of a prior. The theorem is important because it explains why a latent probability model is natural when the observed sequence is exchangeable.

For an infinitely extendable exchangeable sequence of binary variables, there exists a random success probability θ such that, conditional on θ, the variables are iid Bernoulli(θ). In particular,

$$p(y_1, \ldots, y_n) = \int_0^1 \theta^{\sum_i y_i} (1-\theta)^{n-\sum_i y_i} p(\theta)\, \mathrm{d}\theta.$$

This theorem explains why a Bayesian model with a latent probability θ can represent learning from exchangeable binary observations. The theorem requires infinite exchangeability (or infinite extendibility); finite exchangeability alone does not guarantee the exact representation for a fixed finite sequence.

1.7 Prior predictive and posterior predictive distributions

Teacher's spoken explanation

There are two different prediction questions. Before observing data, we average the sampling distribution over the prior to obtain the prior predictive distribution. After observing data, we average over the posterior to obtain the posterior predictive distribution. The first is useful for checking whether the prior model produces plausible data; the second is useful for predicting future observations after learning from the sample.

The **prior predictive distribution** describes data before observing them:

$$p(y) = \int p(y \mid \theta) p(\theta)\, \mathrm{d}\theta.$$

It is useful for understanding the implications of the prior and for checking whether the prior assigns reasonable probability to plausible data.

The **posterior predictive distribution** describes a future observation $\widetilde{Y}$ after observing y:

$$p(\widetilde{y} \mid y) = \int p(\widetilde{y} \mid \theta, y) p(\theta \mid y)\, \mathrm{d}\theta.$$

Under conditional independence, $p(\widetilde{y} \mid \theta, y) = p(\widetilde{y} \mid \theta)$, so

$$p(\widetilde{y} \mid y) = \int p(\widetilde{y} \mid \theta) p(\theta \mid y)\, \mathrm{d}\theta.$$

For a Bernoulli future observation, $\mathbb{P}(\widetilde{Y} = 1 \mid y) = \mathbb{E}(\theta \mid y)$. For a Normal model with known sampling variance σ^2 and posterior $\theta \mid y \sim N(\mu_n, \tau_n^2)$, a new observation has predictive distribution $N(\mu_n, \sigma^2 + \tau_n^2)$. The additional τ_n^2 is uncertainty about the unknown mean.

1.8 Why Bayesian inference? What it does and does not claim

Teacher's spoken explanation

Bayesian inference gives a coherent way to update uncertainty, include prior information, and make probability statements conditional on observed data. But it does not make modelling assumptions disappear. The answer can depend on the likelihood, the prior, the sampling process and the decisions we intend to make. A responsible analysis reports these choices rather than presenting the posterior as an assumption-free truth.

Bayesian analysis can incorporate information available before the current sample, updates beliefs conditional on the observed data, follows the likelihood principle for a fixed prior/model, and produces a full posterior distribution rather than only a point estimate or a repeated-sampling confidence statement. It also permits computational inference for complex models when analytic integration is unavailable.

These advantages do not make the analysis automatic or assumption-free. The result depends on the likelihood, prior, and model structure. Poorly chosen priors, misspecified likelihoods, inappropriate independence assumptions, or weakly identified parameters can produce misleading posteriors. Prior predictive checks, sensitivity analysis, and posterior predictive checks are therefore part of responsible analysis.

1.8.1 Likelihood principle and a classical-inference contrast

Teacher's spoken explanation

The likelihood principle says that, after the data have been observed, likelihood functions proportional as functions of the parameter carry the same inferential information about that parameter. This can differ from procedures whose p-values depend on outcomes that could have occurred but did not. We should understand the contrast without claiming that every classical method is invalid; the point is that the two paradigms condition on information in different ways.

The likelihood principle says that, once the observed data are fixed, all information about the parameter relevant to likelihood-based inference is contained in the likelihood function, up to proportionality. The lecture notes illustrate why two stopping rules can lead to different classical p-values even when the observed likelihoods are proportional: a binomial experiment with a fixed number of tosses and a negative-binomial experiment that stops after a fixed number of successes can yield different tail probabilities for the same observed sequence. Bayesian posterior updating based on the same prior and proportional likelihood gives the same posterior.

A Bayesian credible interval and a frequentist confidence interval also answer different questions. A 95% posterior credible interval has posterior probability 0.95 of containing the parameter, conditional on the observed data and the model. A 95% confidence procedure has 95% coverage over repeated samples under the sampling model; it is not, in the standard frequentist interpretation, a statement that the fixed parameter has 95% probability of lying in the realized interval.

Teacher's spoken explanation

Let us pause and collect what we have built. Bayes' theorem tells us how prior uncertainty is updated by the likelihood. The diagnostic example showed why conditional probabilities must be read in the correct direction. Exchangeability explained why an order-free view of observations can be represented through a latent parameter under the appropriate assumptions. The predictive distribution showed how uncertainty about the parameter becomes uncertainty about a future observation. We have therefore built the basic Bayesian language. Our next task is to make that language calculable for common probability models. That is why we turn to exponential families

and conjugate priors.

Chapter 2

Exponential families and conjugate priors

Teacher's spoken explanation

We now know the logic of updating, but a logic is not yet a practical calculation. Our research team records several kinds of data: yes-or-no outcomes, counts, waiting times, categories and continuous measurements. The likelihood changes with the type of observation. The next question is whether we can choose a prior that combines with each likelihood in a mathematically manageable way. This is the role of conjugacy. We will see that Beta, Gamma, Dirichlet and Normal families are not a list to memorize blindly; each is matched to a likelihood whose algebra has the same kernel structure. By the end of this chapter, you should be able to recognize the pattern and derive the update yourself.

Teacher's spoken explanation

In this chapter we learn a reusable pattern: the likelihood has a recognizable kernel, and a suitably chosen prior keeps the posterior in the same family. We will identify the sufficient statistics and derive the updates model by model.

2.1 Why exponential families matter

Teacher's spoken explanation

Exponential families are important because their likelihoods have a standard algebraic structure. The data enter through a small collection of sufficient statistics, and conjugate priors can often be chosen so that multiplying prior and likelihood leaves us in a known family. This is why the same updating pattern appears repeatedly in Bernoulli, Poisson, Exponential, Normal and Multinomial models.
When you see an exponential-family density, identify the natural parameter, the sufficient statistic and the log-partition function. Those pieces tell you what information from the data matters and often suggest the conjugate prior.

An exponential family is a collection of distributions whose density or mass function can be written in a structured exponential form. One common form is

$$f(y \mid \theta) = h(y) \exp\{\eta(\theta)^{\mathsf{T}} T(y) - A(\theta)\}.$$

Here $T(y)$ is a sufficient-statistic vector, $\eta(\theta)$ is the natural-parameter vector, $A(\theta)$ is the log-

normalizing function, and $h(y)$ is a base measure. For iid observations,

$$f(y_1, \ldots, y_n \mid \theta) = \left\{ \prod_{i=1}^{n} h(y_i) \right\} \exp\left\{ \eta(\theta)^{\mathsf{T}} \sum_{i=1}^{n} T(y_i) - nA(\theta) \right\}.$$

The data enter the likelihood through the sum $\sum_i T(y_i)$. This is why exponential-family models often have simple sufficient statistics and manageable posterior updates.

The lecture notes also use the equivalent representation

$$p(y_i \mid \theta) = f(y_i) g(\theta) \exp\{\phi(\theta)^{\mathsf{T}} u(y_i)\}.$$

In the canonical natural-parameter form, a conjugate prior can be written

$$\pi(\boldsymbol{\eta} \mid \chi, \boldsymbol{\nu}) \propto \exp\{\boldsymbol{\eta}^{\mathsf{T}} \boldsymbol{\nu} - \chi A(\boldsymbol{\eta})\}.$$

For iid data, multiplying by the likelihood gives

$$\pi(\boldsymbol{\eta} \mid y, \chi, \boldsymbol{\nu}) \propto \exp\left\{ \boldsymbol{\eta}^{\mathsf{T}} \left(\boldsymbol{\nu} + \sum_{i=1}^{n} T(y_i) \right) - (\chi + n) A(\boldsymbol{\eta}) \right\}.$$

Thus the conjugate hyperparameters update by adding the observed sufficient-statistic sum to ν and adding the sample size to χ. In the lecture notes' equivalent g, ϕ, u notation, the updated quantities are $\eta + n$ and $\nu + \sum_i u(y_i)$. The exact names and meanings of these hyperparameters depend on the chosen model and parameterization.

Conjugate prior: definition

A prior is conjugate for a likelihood family when the posterior belongs to the same distributional family as the prior. Conjugacy is useful because posterior parameters can often be obtained by adding the observed sufficient statistics to prior hyperparameters. It does not mean that the prior is objectively correct; it is an algebraically convenient family that may or may not be scientifically appropriate.

2.2 Beta–Bernoulli model

Teacher's spoken explanation

Let theta denote the unknown probability of success. A Bernoulli observation contributes one factor of theta for a success and one factor of one minus theta for a failure. Across the sample, the likelihood depends on the data only through the number of successes and failures. A Beta prior has exactly the same powers, so the posterior is again Beta.
This is the simplest example of conjugacy. The prior parameters act like prior success and failure counts, and the data add their actual counts to them.

Let $Y_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, where $0 < \theta < 1$. The Bernoulli mass function is

$$p(y_i \mid \theta) = \theta^{y_i} (1 - \theta)^{1 - y_i}, \qquad y_i \in \{0, 1\}.$$

For $z = \sum_{i=1}^{n} y_i$ successes, the likelihood is

$$L(\theta \mid y) = \theta^z (1 - \theta)^{n - z}.$$

The Beta(α, β) prior has density

$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1}, \quad 0 < \theta < 1,$$

where $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$.

Multiplication gives

$$p(\theta \mid y) \propto \theta^{\alpha+z-1}(1-\theta)^{\beta+n-z-1},$$

so

Beta–Bernoulli posterior

$$\boxed{\theta \mid y \sim \text{Beta}(\alpha+z,\ \beta+n-z).}$$

The prior contributes $\alpha-1$ and $\beta-1$ to the exponents; the data contribute z successes and $n-z$ failures.

For the posterior $\text{Beta}(a_n, b_n)$,

$$\mathbb{E}(\theta \mid y) = \frac{a_n}{a_n+b_n}, \qquad \text{Var}(\theta \mid y) = \frac{a_n b_n}{(a_n+b_n)^2(a_n+b_n+1)}.$$

If $a_n, b_n > 1$, the posterior mode is $(a_n-1)/(a_n+b_n-2)$. If either parameter is at most one, the mode may occur at a boundary; do not use the interior formula blindly.

2.2.1 Uniform prior and Laplace's rule of succession

Teacher's spoken explanation

A Uniform zero-to-one prior is a Beta prior with both shape parameters equal to one. After observing z successes in n Bernoulli trials, the posterior is Beta with parameters z plus one and n minus z plus one. The predictive probability of a success on the next trial is therefore z plus one divided by n plus two.
This is Laplace's rule of succession. It avoids assigning exactly zero probability to a future success merely because no success has yet been observed in a small sample.

A uniform prior on $(0,1)$ is $\text{Beta}(1,1)$. The posterior becomes

$$\theta \mid y \sim \text{Beta}(z+1, n-z+1), \qquad \mathbb{P}(\widetilde{Y}=1 \mid y) = \mathbb{E}(\theta \mid y) = \frac{z+1}{n+2}.$$

The quantity $(z+1)/(n+2)$ is Laplace's rule of succession. It differs from the maximum-likelihood estimate z/n by adding one prior pseudo-success and one prior pseudo-failure. As n grows with a stable success proportion, the difference becomes small.

Under the uniform prior, the prior predictive distribution of the count $Z = \sum_i Y_i$ is

$$\mathbb{P}(Z=z) = \binom{n}{z}\int_0^1 \theta^z(1-\theta)^{n-z}\,\mathrm{d}\theta = \binom{n}{z}B(z+1, n-z+1) = \frac{1}{n+1}.$$

Thus the count Z is discrete uniform on $\{0, 1, \ldots, n\}$. This statement concerns the count, not the probability of every ordered binary sequence.

2.2.2 Beta–Binomial form

Teacher's spoken explanation

The Bernoulli model records each trial separately, while the Binomial model records the total number of successes. The likelihood kernel in theta is the same, so the Beta prior remains conjugate. The combinatorial coefficient affects the marginal likelihood of the count but not the posterior shape as a function of theta.

If the observed data are summarized as $Z \mid \theta \sim \text{Binomial}(n, \theta)$, the same Beta prior gives the same posterior. The prior predictive distribution of the count is Beta–Binomial:

$$\mathbb{P}(Z = z) = \binom{n}{z} \frac{B(\alpha + z, \beta + n - z)}{B(\alpha, \beta)}, \qquad z = 0, \ldots, n.$$

For m new trials after observing z successes in n trials, the number K of future successes has

$$\mathbb{P}(K = k \mid y) = \binom{m}{k} \frac{B(\alpha + z + k, \beta + n - z + m - k)}{B(\alpha + z, \beta + n - z)}.$$

2.3 Gamma–Poisson model

Teacher's spoken explanation

For Poisson data, the likelihood contains a power of lambda determined by the total count and an exponential term determined by the total exposure. A Gamma prior uses the same power-exponential form. With a shape-rate Gamma convention, counts increase the shape and exposure increases the rate.
Always check whether the observation interval has length one. When exposure differs across observations, the total exposure must appear in the posterior rate; omitting it is a frequent source of incorrect answers.

Use the Gamma shape–rate parameterization throughout this guide:

$$\lambda \sim \text{Gamma}(a, b), \qquad p(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}, \quad \lambda > 0.$$

The mean is a/b and the variance is a/b^2. The second parameter is a **rate**, not a scale. If a source uses a Gamma scale parameter, the formulas look different.

Suppose $Y_i \mid \lambda \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$. The likelihood kernel is

$$L(\lambda \mid y) \propto \lambda^{\sum_i y_i} e^{-n\lambda}.$$

Multiplying by the Gamma prior gives

$$p(\lambda \mid y) \propto \lambda^{a + \sum_i y_i - 1} e^{-(b+n)\lambda}.$$

Gamma–Poisson update

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + \sum_{i=1}^{n} y_i,\ b + n\right).}$$

With exposure times t_i and $Y_i \mid \lambda \sim \text{Poisson}(t_i \lambda)$, replace n by $\sum_i t_i$ in the posterior rate:

$$\lambda \mid y \sim \text{Gamma}\left(a + \sum_i y_i,\ b + \sum_i t_i\right).$$

For a new count with unit exposure, the posterior predictive distribution is negative-binomial:

$$\mathbb{P}(\widetilde{Y} = k \mid y) = \frac{\Gamma(a_n + k)}{\Gamma(a_n) k!} \left(\frac{b_n}{b_n + 1}\right)^{a_n} \left(\frac{1}{b_n + 1}\right)^{k},$$

where $a_n = a + \sum_i y_i$ and $b_n = b + n$. Its predictive mean is a_n / b_n, equal to the posterior mean of λ.

2.4 Gamma–Exponential model

Teacher's spoken explanation

For Exponential observations parameterized by a rate lambda, the likelihood is proportional to lambda raised to the sample size multiplied by the exponential of minus lambda times the sum of the observations. A Gamma prior is conjugate, so the posterior shape increases by the sample size and the posterior rate increases by the sum of the observations.
Notice the difference from the Poisson exposure example: for Exponential waiting times the observed waiting-time sum contributes to the rate parameter, whereas the number of observations contributes to the shape.

Let $Y_i \mid \lambda \overset{\text{iid}}{\sim} \text{Exponential}(\lambda)$ with **rate** λ, so

$$f(y_i \mid \lambda) = \lambda e^{-\lambda y_i}, \qquad y_i > 0.$$

For n observations, the likelihood kernel is $\lambda^n e^{-\lambda \sum_i y_i}$. With $\lambda \sim \text{Gamma}(a, b)$,

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + n,\ b + \sum_{i=1}^{n} y_i\right).}$$

The posterior mean is $(a+n)/(b+\sum_i y_i)$. The data update the shape by the number of observations and the rate by the total observed waiting time. This interpretation is useful: many short waiting times support a larger rate; a large total waiting time supports a smaller rate.

Exam check

Always identify whether the Exponential parameter is a rate or a mean. If the parameter is the mean $\vartheta = 1/\lambda$, the likelihood and conjugate prior are expressed differently. The Gamma conjugacy above is for the rate λ.

2.5 Dirichlet–Multinomial model

Teacher's spoken explanation

The Multinomial model has a vector of category probabilities that sum to one. The Dirichlet prior is the natural multivariate extension of the Beta prior. Each category count is added to its corresponding Dirichlet parameter. The posterior remains Dirichlet, and the posterior predictive probability for a category is its updated parameter divided by the total updated concentration.

Suppose each observation falls into one of K categories with probabilities $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)$, where $\theta_k \geq 0$ and $\sum_k \theta_k = 1$. The likelihood for counts $n_1, \ldots, n_K$ (with $n = \sum_k n_k$) is

$$p(n_1, \ldots, n_K \mid \boldsymbol{\theta}) = \frac{n!}{\prod_{k=1}^{K} n_k!} \prod_{k=1}^{K} \theta_k^{n_k}.$$

A Dirichlet prior has density on the simplex

$$p(\boldsymbol{\theta}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^{K} \theta_k^{\alpha_k - 1}, \qquad B(\boldsymbol{\alpha}) = \frac{\prod_k \Gamma(\alpha_k)}{\Gamma(\alpha_0)}, \quad \alpha_0 = \sum_k \alpha_k.$$

The posterior is

Dirichlet–Multinomial update

$$\boxed{\boldsymbol{\theta} \mid y \sim \text{Dirichlet}(\alpha_1 + n_1, \ldots, \alpha_K + n_K).}$$

The posterior predictive probability that the next observation belongs to category k is

$$\mathbb{P}(\widetilde{Y} = k \mid y) = \frac{\alpha_k + n_k}{\alpha_0 + n}.$$

The Dirichlet distribution is the multi-category analogue of the Beta distribution. For $K = 2$, a Dirichlet(α_1, α_2) distribution reduces to a Beta(α_1, α_2) distribution for the first category probability. The Dirichlet–Multinomial prior predictive distribution is

$$p(n_1, \ldots, n_K) = \frac{n!}{\prod_k n_k!} \frac{B(\alpha_1 + n_1, \ldots, \alpha_K + n_K)}{B(\alpha_1, \ldots, \alpha_K)}.$$

2.6 Normal likelihood with known variance: Normal–Normal conjugacy

Teacher's spoken explanation

Here the unknown quantity is a Normal mean, while the sampling variance is treated as known. A Normal prior is conjugate because multiplying two Gaussian kernels in the unknown mean produces another Gaussian kernel. The posterior precision is the sum of prior precision and data precision.
The posterior mean is a precision-weighted average of the prior mean and sample mean. A more precise source receives more weight. This precision interpretation is usually clearer than memorizing the formula alone.

Suppose $Y_i \mid \mu \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, with known σ^2, and prior $\mu \sim N(\mu_0, \tau_0^2)$. The sample mean satisfies $\overline{Y} \mid \mu \sim N(\mu, \sigma^2/n)$. The posterior variance and mean are

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1}, \qquad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\overline{y}}{\sigma^2}\right).$$

Therefore

Normal–Normal posterior

$$\boxed{\mu \mid y \sim N(\mu_n, \tau_n^2).}$$

The posterior mean is a precision-weighted average:

$$\mu_n = \frac{\mu_0/\tau_0^2 + \overline{y}/(\sigma^2/n)}{1/\tau_0^2 + 1/(\sigma^2/n)}.$$

Precision means inverse variance. A component with smaller variance has greater precision and receives greater weight.

As n increases, the data precision n/σ^2 increases, so the posterior mean is pulled toward $\overline{y}$ and posterior variance decreases. As $\tau_0^2 \to \infty$, the prior becomes diffuse and the posterior approaches $N(\overline{y}, \sigma^2/n)$. As $\tau_0^2 \to 0$, the prior concentrates at μ_0 and dominates the data.

For a new observation,

$$\widetilde{Y} \mid y \sim N(\mu_n, \sigma^2 + \tau_n^2).$$

Do not confuse the posterior distribution of the unknown mean (variance τ_n^2) with the predictive distribution of a new observation (variance $\sigma^2 + \tau_n^2$).

2.7 Unknown Normal mean and variance: Normal–Inverse-Gamma

Teacher's spoken explanation

When both the mean and variance are unknown, we need a joint prior. The Normal–Inverse-Gamma construction gives a Normal conditional prior for the mean given the variance and an inverse-Gamma prior for the variance. The posterior updates several hyperparameters using the sample size, sample mean and residual sum of squares.
The key is to keep the variance and precision parameterizations separate. The inverse-Gamma formulation is convenient for variance; the Normal–Gamma formulation is convenient when working with precision.

The lecture notes use scaled-inverse-chi-square distributions and also motivate the joint prior proportional to $1/\sigma^2$. A convenient equivalent conjugate formulation is the Normal–Inverse-Gamma (NIG) prior. Use the inverse-Gamma shape–scale convention

$$\sigma^2 \sim \text{InvGamma}(a_0, b_0), \qquad p(\sigma^2) = \frac{b_0^{a_0}}{\Gamma(a_0)}(\sigma^2)^{-a_0-1}\exp\left(-\frac{b_0}{\sigma^2}\right),$$

and conditionally

$$\mu \mid \sigma^2 \sim N\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right).$$

This is a Normal–Gamma prior if parameterized in terms of precision $\lambda = 1/\sigma^2$: $\lambda \sim \text{Gamma}(a_0, b_0)$ and $\mu \mid \lambda \sim N(\mu_0, 1/(\kappa_0\lambda))$.

For $Y_i \mid \mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, define $\overline{y} = n^{-1}\sum_i y_i$ and $S = \sum_i (y_i - \overline{y})^2$. The posterior hyperparameters are

$$\kappa_n = \kappa_0 + n, \qquad \mu_n = \frac{\kappa_0\mu_0 + n\overline{y}}{\kappa_n},$$

$$a_n = a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{S}{2} + \frac{\kappa_0 n}{2\kappa_n}(\overline{y} - \mu_0)^2.$$

Thus

$$\boxed{\sigma^2 \mid y \sim \text{InvGamma}(a_n, b_n), \qquad \mu \mid \sigma^2, y \sim N\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right).}$$

The posterior mean of σ^2 exists only if $a_n > 1$ and is $b_n/(a_n - 1)$. The posterior predictive distribution is Student's t:

$$\boxed{\widetilde{Y} \mid y \sim t_{2a_n}\left(\mu_n, \sqrt{\frac{b_n(\kappa_n + 1)}{a_n\kappa_n}}\right),}$$

where the second argument is the Student-t scale, not its standard deviation. The variance, when $a_n > 1$, is the squared scale multiplied by $2a_n/(2a_n - 2)$.

2.7.1 Normal–Gamma precision parameterization: the form named in the syllabus

Teacher's spoken explanation

The syllabus names the Normal–Gamma prior, which is most naturally expressed using precision tau equal to one over sigma squared. Conditional on tau, the mean is Normal, and tau has a Gamma prior. The likelihood contributes both a quadratic term in the mean and a power of the precision. Completing the square identifies the updated Normal conditional distribution and updated Gamma marginal distribution.

The same conjugate model is often written using the precision $\lambda = 1/\sigma^2$ instead of the variance. The Gamma distribution is then placed directly on precision:

$$\lambda \sim \text{Gamma}(a_0, b_0), \qquad \mu \mid \lambda \sim N\left(\mu_0, \frac{1}{\kappa_0 \lambda}\right), \qquad Y_i \mid \mu, \lambda \overset{\text{iid}}{\sim} N\left(\mu, \frac{1}{\lambda}\right).$$

Here $\text{Gamma}(a_0, b_0)$ uses shape a_0 and rate b_0. The posterior is

$$\lambda \mid y \sim \text{Gamma}(a_n, b_n), \qquad \mu \mid \lambda, y \sim N\left(\mu_n, \frac{1}{\kappa_n \lambda}\right),$$

with

$$\kappa_n = \kappa_0 + n, \qquad \mu_n = \frac{\kappa_0 \mu_0 + n\bar{y}}{\kappa_0 + n},$$

$$a_n = a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{1}{2}\sum_{i=1}^{n}(y_i - \bar{y})^2 + \frac{1}{2}\frac{\kappa_0 n}{\kappa_0 + n}(\bar{y} - \mu_0)^2.$$

The update follows by writing the joint prior-times-likelihood kernel and completing the square in μ. The term $\sum_i (y_i - \bar{y})^2$ measures within-sample variation; the final term accounts for the difference between the prior centre and the sample mean. This precision form and the Normal–Inverse-Gamma variance form above are equivalent under $\lambda = 1/\sigma^2$ and consistent transformation of the prior density. Do not treat a Gamma prior on variance as interchangeable with a Gamma prior on precision.

2.7.2 Scaled-inverse-chi-square form used in the lecture notes

Teacher's spoken explanation

The scaled-inverse-chi-square distribution is another way to express a prior for a Normal variance. It is closely related to the inverse-Gamma distribution. Our task in an exam is not to confuse the names, but to identify the parameterization and then use the matching update formulas consistently.

A random variable V has a scaled-inverse-chi-square distribution with degrees of freedom ν and scale s^2, written $V \sim \text{Inv-}\chi^2(\nu, s^2)$, when

$$\frac{\nu s^2}{V} \sim \chi^2_\nu.$$

This is equivalent to $V \sim \text{InvGamma}(\nu/2, \nu s^2/2)$. If $Y_i \mid \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with known μ and prior $\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, s_0^2)$, then

$$\nu_n = \nu_0 + n, \qquad s_n^2 = \frac{\nu_0 s_0^2 + \sum_i (y_i - \mu)^2}{\nu_0 + n},$$

$$\sigma^2 \mid y \sim \text{Inv-}\chi^2(\nu_n, s_n^2), \qquad \frac{\widetilde{Y} - \mu}{s_n} \mid y \sim t_{\nu_n}.$$

The last statement is a posterior predictive result obtained by integrating a Normal likelihood over the inverse-chi-square posterior for the variance.

2.8 Conjugacy summary table

Teacher's spoken explanation

Before moving on, let us compare the conjugate pairs in one place. The underlying pattern is always the same: choose a prior whose kernel has the same functional form as the likelihood in the unknown parameter. Multiplication then changes the hyperparameters rather than changing

the family.

Likelihood	Prior	Posterior
Bernoulli(θ), z successes in n	$\theta \sim \text{Beta}(\alpha, \beta)$	$\theta \mid y \sim \text{Beta}(\alpha + z, \beta + n - z)$
Binomial(n, θ)	$\theta \sim \text{Beta}(\alpha, \beta)$	$\theta \mid z \sim \text{Beta}(\alpha + z, \beta + n - z)$
Poisson(λ), n unit exposures	$\lambda \sim \text{Gamma}(a, b)$ (rate)	$\lambda \mid y \sim \text{Gamma}(a + \sum y_i, b + n)$
Poisson($t_i\lambda$)	$\lambda \sim \text{Gamma}(a, b)$ (rate)	$\lambda \mid y \sim \text{Gamma}(a + \sum y_i, b + \sum t_i)$
Exponential rate λ	$\lambda \sim \text{Gamma}(a, b)$ (rate)	$\lambda \mid y \sim \text{Gamma}(a + n, b + \sum y_i)$
Multinomial counts $\boldsymbol{\theta}$	$\boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\alpha})$	$\boldsymbol{\theta} \mid y \sim \text{Dirichlet}(\boldsymbol{\alpha} + \boldsymbol{n})$
Normal mean μ, known σ^2	$\mu \sim N(\mu_0, \tau_0^2)$	$\mu \mid y \sim N(\mu_n, \tau_n^2)$
Normal (μ, σ^2) both unknown	$\sigma^2 \sim \text{InvGamma}(a_0, b_0)$, $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0)$	NIG posterior with $(\mu_n, \kappa_n, a_n, b_n)$ above

Exam check

The easiest way to identify a conjugate posterior is to compare powers and exponential terms. Match the posterior kernel to a known density, then read off the updated parameters. Do not memorize only the final update: be able to show the multiplication that creates it.

Teacher's spoken explanation

The important achievement of this chapter is not memorizing five posterior formulas. It is learning to recognize a repeated algebraic structure. In each model, the likelihood contributes information through a sufficient statistic, and the conjugate prior contributes prior information through matching powers or quadratic terms. The posterior retains the same family, with updated parameters that combine prior information and sample information. We now possess a posterior distribution for many common problems. The next question is what to report from it and how to predict. That takes us naturally to Bayesian estimation.

Chapter 3

Bayesian estimation, MAP and credible intervals

Teacher's spoken explanation

At this point our team has a posterior distribution. Have we answered the scientific question? We have answered it in a rich sense, because we have described uncertainty over all plausible parameter values. But a report may still ask for one number, or for a range of plausible values. That is the problem of Bayesian estimation. We now have to decide how to summarize a distribution without pretending that a single summary contains all its information. Posterior mean, median and mode answer slightly different questions; credible intervals describe ranges; and posterior prediction shifts attention from an unknown parameter to a future observation. The key principle is that an estimate is a decision or summary extracted from the posterior, not a replacement for the posterior itself.

Teacher's spoken explanation

This chapter asks how we summarize a posterior distribution. There is no single estimator that is best for every purpose. We connect posterior summaries to loss functions, then distinguish credible intervals from confidence intervals and explain predictive uncertainty.

3.1 Point estimation from the posterior

Teacher's spoken explanation

A posterior distribution contains more information than a single estimate. If a question asks for a point estimate, we must decide which summary is appropriate. The posterior mean, median and mode are not automatically identical; each is optimal under a different loss function or criterion. Always read the wording of the question before choosing one.

A posterior distribution contains more information than a single estimate. A point estimate is a decision extracted from that distribution. The appropriate point estimate depends on the loss function.

Loss function	Posterior-optimal estimate	Reason
Squared-error loss $L(a,\theta) = (a-\theta)^2$	Posterior mean $\mathbb{E}(\theta \mid y)$	It minimizes posterior expected squared error.
Absolute-error loss $L(a,\theta) = \lvert a-\theta \rvert$	A posterior median	It minimizes posterior expected absolute error.

Zero–one loss (small neighborhoods)	Posterior mode / MAP	The mode maximizes posterior density or mass.
Asymmetric absolute/quantile loss	A posterior quantile	The desired quantile reflects the relative costs of over- and underestimation.

3.1.1 Posterior mean

Teacher's spoken explanation

Under squared-error loss, the posterior mean minimizes posterior expected loss. It averages over the entire posterior distribution, so tail probability can affect it. For a conjugate distribution, we often obtain it from a standard mean formula.

The posterior mean is

$$\widehat{\theta}_{\mathrm{PM}} = \mathbb{E}(\theta \mid y) = \int \theta p(\theta \mid y)\,\mathrm{d}\theta.$$

It is particularly natural under squared-error loss. It can lie in a region of relatively low posterior density for a multimodal posterior, so a plot should be inspected before using it as a complete summary.

3.1.2 Posterior median

Teacher's spoken explanation

Under absolute-error loss, the posterior median is a natural optimal estimate. It divides the posterior probability into two equal halves. Unlike the mean, it is less sensitive to the magnitude of extreme values in the tails.

A posterior median m satisfies

$$\mathbb{P}(\theta \le m \mid y) \ge \frac{1}{2}, \qquad \mathbb{P}(\theta \ge m \mid y) \ge \frac{1}{2}.$$

For a continuous posterior, $F(m \mid y) = 1/2$. A median is often more robust to skewness and outliers than the mean.

3.1.3 Posterior mode and maximum a posteriori estimate

Teacher's spoken explanation

The posterior mode is the value at which the posterior density is highest. It is called the MAP estimate. Since the posterior is proportional to prior times likelihood, maximizing the posterior is equivalent to maximizing the log prior plus the log likelihood. This shows directly how the prior can shift an estimate away from the maximum likelihood estimate.

The MAP estimate maximizes the posterior density:

$$\widehat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(\theta \mid y) = \arg\max_{\theta} \{p(y \mid \theta)p(\theta)\}.$$

Equivalently, it maximizes the log posterior. Since $p(y)$ is constant in θ, it can be omitted. The likelihood-only maximizer is the MLE:

$$\widehat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} p(y \mid \theta).$$

If the prior is constant over the parameter space and the maximum is interior, MAP and MLE coincide. With an informative prior they generally differ.

MAP is not the same as MLE

MLE uses only the likelihood. MAP uses likelihood multiplied by the prior. A prior can pull the MAP estimate away from the MLE. Also, unlike the MLE, the MAP is generally not invariant under a one-to-one reparameterization because a density changes by a Jacobian factor.

3.1.4 Why MAP is not invariant under transformations

Teacher's spoken explanation

A density is defined relative to a coordinate system. When we transform a random variable, a Jacobian factor changes its density. Therefore the location of the maximum density need not transform as simply as the random variable itself. This is why MAP estimates are not generally invariant under nonlinear transformations.

Let $\psi = g(\theta)$ be a differentiable one-to-one transformation. The transformed posterior density is

$$p_\psi(\psi \mid y) = p_\theta(\theta \mid y) \left| \frac{\mathrm{d}\theta}{\mathrm{d}\psi} \right|.$$

The Jacobian can change the location of the maximum. Therefore, the mode of the density of ψ need not be g of the mode of the density of θ. The MLE is invariant under one-to-one transformations because maximizing the likelihood over θ is equivalent to maximizing the same likelihood after relabeling the parameter; the density transformation argument does not apply to the likelihood as a density over the parameter.

3.2 Credible intervals

Teacher's spoken explanation

A Bayesian credible interval is a range that contains a specified amount of posterior probability, conditional on the observed data and the model. A 95 percent credible interval means that the posterior probability of the parameter lying in the interval is 0.95. This is a direct probability statement about the parameter under the posterior.

A posterior credible set $C(y)$ with credibility level $1 - \alpha$ satisfies

$$\mathbb{P}(\theta \in C(y) \mid y) = 1 - \alpha.$$

For a continuous scalar parameter, common choices are equal-tail intervals and highest-posterior-density (HPD) intervals.

3.2.1 Equal-tail interval

Teacher's spoken explanation

An equal-tail interval leaves the same posterior probability in each tail. For a 95 percent interval, 2.5 percent is below the lower endpoint and 2.5 percent is above the upper endpoint. Its endpoints are obtained from posterior quantiles.

The equal-tail interval leaves posterior probability $\alpha/2$ below and $\alpha/2$ above:

$$C_{ET} = \left[q_{\alpha/2}, q_{1-\alpha/2} \right], \quad F(q_u \mid y) = u.$$

For a 95% interval, use posterior quantiles 0.025 and 0.975. For a 90% interval, use 0.05 and 0.95. This is the convention to state explicitly in an examination answer.

3.2.2 Highest-posterior-density interval

Teacher's spoken explanation

An HPD interval contains the specified posterior probability while including points with the highest density first. For a unimodal density it is often the shortest interval with that probability. Unlike an equal-tail interval, its tail probabilities need not be equal.

An HPD region contains posterior mass $1 - \alpha$ and is made up of points with the highest posterior density. For a unimodal continuous posterior, its endpoints typically have equal posterior density. HPD intervals are shortest among intervals for a one-dimensional unimodal density under regular conditions, but can be disconnected for a multimodal posterior. Equal-tail and HPD intervals need not agree, especially for skewed distributions.

Beta posterior formulas frequently used for credible intervals

If $\theta \mid y \sim \text{Beta}(a, b)$, then

$$F_\theta(x \mid y) = I_x(a, b), \qquad q_u = I_u^{-1}(a, b),$$

where I_x is the regularized incomplete Beta function. There is no generally simple elementary formula for these quantiles; they are obtained from a Beta quantile function or numerical integration/inversion. In R, use `qbeta(u, a, b)`.

3.3 Credible interval versus confidence interval

Teacher's spoken explanation

Please do not interchange these two interval interpretations. A confidence procedure has a repeated-sampling coverage property: across repetitions, the procedure covers the fixed parameter at the stated rate. A Bayesian credible interval has a posterior probability interpretation after the data have been observed. The numerical intervals may sometimes look similar, but the interpretations differ.

A Bayesian credible interval is conditional on the observed data and prior/model: the posterior probability that the parameter is in the interval is the stated level. A frequentist confidence interval is a random interval generated by a procedure whose coverage probability under repeated sampling is the stated level. The lecture notes highlight that confidence procedures can have unintuitive behavior in some models (for example, a uniform-location model) and that their meaning is not the same as a posterior probability statement.

3.4 Posterior prediction and predictive uncertainty

Teacher's spoken explanation

To predict a new observation, we must account for uncertainty in the parameter as well as the randomness in a future observation conditional on that parameter. We do this by integrating the sampling distribution over the posterior. This integration is why the predictive distribution is often more dispersed than a plug-in distribution that substitutes one estimate.

For a future observation $\widetilde{Y}$,

$$p(\widetilde{y} \mid y) = \int p(\widetilde{y} \mid \theta) p(\theta \mid y)\, \mathrm{d}\theta.$$

This distribution accounts for both (i) sampling variability in a future observation conditional on θ and (ii) posterior uncertainty about θ. In many models the predictive distribution is wider than the posterior distribution for a parameter for precisely this reason.

For the Beta–Bernoulli model,

$$\mathbb{P}(\widetilde{Y} = 1 \mid y) = \frac{\alpha + z}{\alpha + \beta + n}.$$

For the Normal–Normal model with known sampling variance,

$$\widetilde{Y} \mid y \sim N(\mu_n, \sigma^2 + \tau_n^2).$$

For the NIG Normal model, the posterior predictive distribution is Student-t with the parameters given in Chapter 2.

Exam check

When asked for a credible interval, interval estimation, or posterior probability, do not report a classical confidence interval unless the question explicitly asks for one. State whether the interval is equal-tail or HPD, identify the posterior distribution first, and then identify the correct posterior quantiles.

Teacher's spoken explanation

We have learned that a posterior distribution can be summarized in several ways, and that the appropriate summary depends on the question and the loss function. We have also distinguished uncertainty about a parameter from uncertainty about a future observation. But sometimes the scientific question is not asking for a value or interval at all. It asks whether one explanation is more credible than another. To answer that, we must compare hypotheses using the posterior odds and the Bayes factor. So the next chapter changes the target of inference while retaining the same Bayesian updating principle.

Chapter 4

Bayesian hypothesis testing and Bayes factors

Teacher's spoken explanation

So far, we have mostly asked, 'What values of the parameter remain plausible?' Scientists and decision-makers often ask a different question: 'Which of two explanations is better supported?' For example, does the data support a baseline recovery probability, or a different one? Bayesian hypothesis testing answers this through posterior odds. The Bayes factor measures how much the observed data change the odds by comparing their predictive support under the competing hypotheses. Notice the continuity: the evidence in Bayes' theorem, which we first treated as a normalizing constant, now becomes the central object in model comparison.

Teacher's spoken explanation

This chapter develops Bayesian evidence for competing hypotheses. We move from simple hypotheses to composite hypotheses, where uncertainty within each model must be averaged out. Posterior odds and the Bayes factor are our main tools.

4.1 Testing hypotheses in the Bayesian framework

Teacher's spoken explanation

Bayesian hypothesis testing compares the posterior support for competing hypotheses. The central output is often posterior odds. The data change prior odds through a Bayes factor, which measures the evidence contributed by the observed data.

A Bayesian hypothesis test compares hypotheses using posterior probabilities. For two hypotheses,

$$\mathbb{P}(H_0 \mid y) + \mathbb{P}(H_1 \mid y) = 1.$$

A decision rule can choose the hypothesis with larger posterior probability, or use a different threshold if the costs of the two kinds of errors are different. The posterior probability itself is not the same thing as a frequentist p-value.

4.1.1 Simple hypotheses

Teacher's spoken explanation

For two simple hypotheses, each hypothesis fixes the parameter at a specified value. We evaluate the likelihood of the observed data under each value and take their ratio. That ratio is the Bayes factor. It multiplies prior odds to produce posterior odds.

A simple hypothesis specifies the parameter completely, for example $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$. Then

$$\mathrm{BF}_{10} = \frac{p(y \mid H_1)}{p(y \mid H_0)}$$

is the likelihood ratio for the two specified models, and

$$\frac{\mathbb{P}(H_1 \mid y)}{\mathbb{P}(H_0 \mid y)} = \mathrm{BF}_{10}\frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)}.$$

4.1.2 Composite hypotheses and marginal likelihood

Teacher's spoken explanation

A composite hypothesis allows a range of parameter values. We cannot evaluate it at one arbitrary value. Instead, we average the likelihood over the prior distribution within that hypothesis. The result is the marginal likelihood, and the ratio of two marginal likelihoods is the Bayes factor.

A composite hypothesis contains an unknown parameter with a prior under that hypothesis. For H_j with parameter θ_j,

$$m_j(y) = p(y \mid H_j) = \int p(y \mid \theta_j, H_j)\pi_j(\theta_j \mid H_j)\,\mathrm{d}\theta_j.$$

The quantity $m_j(y)$ is the marginal likelihood (also called model evidence or prior predictive density under model j). The Bayes factor is

$$\boxed{\mathrm{BF}_{10} = \frac{m_1(y)}{m_0(y)}.}$$

The parameter prior within each composite hypothesis is essential: the Bayes factor averages the likelihood over that prior. It is not generally correct to substitute the likelihood at the MLE into the Bayes factor.

4.2 Interpreting posterior odds and Bayes factors

Teacher's spoken explanation

A Bayes factor of, say, six in favour of one hypothesis means the observed data are six times as probable under that hypothesis as under the competing one, in the simple-hypothesis setting. It is not itself a posterior probability. To obtain posterior probability, combine the Bayes factor with prior odds and then normalize.

If $\mathrm{BF}_{10} = 5$, the data multiply the prior odds in favor of H_1 by five. It does not follow that the posterior probability of H_1 is $5/6$ unless the prior model probabilities are equal. The posterior probability is

$$\mathbb{P}(H_1 \mid y) = \frac{\mathrm{BF}_{10}\mathbb{P}(H_1)}{\mathrm{BF}_{10}\mathbb{P}(H_1) + \mathbb{P}(H_0)}.$$

The Bayes factor compares the data compatibility of the models. The prior odds describe the prior plausibility of the models. Both are required for posterior odds.

A common qualitative convention is to describe Bayes factors close to one as weak evidence, and increasingly large or small values as stronger evidence. Numerical labels such as "substantial" or "strong" are conventions, not universal mathematical thresholds; always state which direction the ratio uses. This guide uses BF_{10}, so values above one favor H_1.

4.3 Worked disease-test Bayes factor

Teacher's spoken explanation

We revisit diagnostic testing from the perspective of hypothesis comparison. The likelihood ratio tells us how much a positive result favours disease over no disease. The low prior prevalence still matters, so we must not confuse strong relative evidence with a high final probability.

In the diagnostic example, take $H_1 = D$ and $H_0 = D^c$. The Bayes factor for a positive test is

$$\mathrm{BF}_{10} = \frac{\mathbb{P}(T^+ \mid D)}{\mathbb{P}(T^+ \mid D^c)} = \frac{0.95}{0.10} = 9.5.$$

The prior odds are $0.02/0.98 = 1/49$. Therefore posterior odds are $9.5/49 \approx 0.19388$, and posterior disease probability is $0.19388/(1+0.19388) = 0.1624$. The Bayes factor is 9.5 even though the posterior disease probability is only about 0.1624, because the prior probability of disease is small.

4.4 Bayesian testing versus classical testing

Teacher's spoken explanation

A classical test often asks how surprising the observed statistic, or something more extreme, would be under the null hypothesis. A Bayes factor instead compares the probability of the observed data under competing models, averaging over parameter uncertainty when a model is composite. These are different questions, and their numerical outputs should not be interpreted interchangeably.

A frequentist test controls a long-run Type I error probability under a specified null sampling model and reports a p-value or rejection decision. A Bayesian test computes posterior model probabilities or odds, which depend on model priors and, for composite hypotheses, parameter priors. These are different procedures and answer different questions.

Exam check

For a hypothesis-testing calculation, write the four quantities in this order: prior model odds, marginal likelihood under H_0, marginal likelihood under H_1, and Bayes factor. Only then calculate posterior odds or posterior model probabilities. This prevents the common mistake of treating the Bayes factor as the posterior probability.

Teacher's spoken explanation

We have seen that the Bayes factor is a ratio of marginal likelihoods, and that a composite hypothesis requires us to average its likelihood over the prior distribution within that model. This means the prior is not merely a starting point for estimation; it is part of the predictive definition of a model. We cannot discuss Bayesian testing responsibly without discussing prior choice. Our next chapter therefore steps back from the test itself and examines how priors are selected, when they are proper, and how Jeffreys' construction uses Fisher information.

Chapter 5

Choice of priors: conjugate, flat, diffuse and Jeffreys

Teacher's spoken explanation

We have already used priors to update parameters and to compare hypotheses. It is now time to ask a question that we deliberately postponed: how should the prior be chosen? This is not a cosmetic decision. In parameter estimation it influences the posterior, and in model comparison it can influence the marginal likelihood substantially. We will distinguish a conjugate prior, a proper prior, an improper prior, a diffuse prior and a non-informative construction. Jeffreys' prior gives us a principled route based on Fisher information and invariance under reparameterization. The lesson is not that one prior is always correct; it is that prior assumptions must be explicit, mathematically valid and examined for sensitivity.

Teacher's spoken explanation

A prior is a modelling choice that must be explained. We compare conjugate, flat, diffuse and Jeffreys constructions, distinguish proper from improper priors, and learn to examine sensitivity.

5.1 How should a prior be chosen?

Teacher's spoken explanation

A prior should represent relevant knowledge before the current data are used. It may come from scientific information, previous studies, a conjugate family chosen for computational convenience, or a carefully justified weakly informative construction. We should explain the prior's meaning and examine how sensitive important conclusions are to reasonable alternatives.

A prior should represent defensible information or a transparent regularization choice before seeing the current data. In practical modeling, the prior should be assessed through its implications for plausible parameter values and plausible data. The following terms are related but not interchangeable.

- **Conjugate prior:** chosen so the posterior remains in the same family as the prior. The main benefit is algebraic simplicity.
- **Flat prior:** constant with respect to a particular parameterization, such as $\pi(\theta) \propto 1$. Flatness depends on the coordinate used and is not automatically non-informative.
- **Diffuse prior:** broad relative to a scientifically meaningful scale. A diffuse proper prior still integrates to one.
- **Non-informative/reference prior:** a prior constructed to contribute relatively little information under a stated criterion. There is no single universally non-informative prior for all models and

parameterizations.

- **Jeffreys' prior:** a rule based on Fisher information that is invariant under smooth one-to-one reparameterization in the one-parameter case.

5.2 Proper and improper priors

Teacher's spoken explanation

A proper prior integrates to one. An improper prior does not, but may sometimes produce a proper posterior after observing data. An improper prior cannot be treated as an ordinary probability distribution for prior prediction or model comparison without additional care. Always check whether the posterior is proper before reporting an inference.

A proper prior is a probability density or mass function that integrates or sums to one. An improper prior is a nonnegative kernel that has infinite total integral, for example $\pi(\theta) \propto 1$ on the whole real line. An improper prior cannot be interpreted directly as a probability distribution, but it can sometimes yield a proper posterior after multiplication by the likelihood.

If $\pi(\theta)$ is an improper prior kernel, the posterior is proper only if

$$0 < \int \pi(\theta) p(y \mid \theta)\, \mathrm{d}\theta < \infty.$$

The finite normalizing integral must be checked. It is not enough to write the posterior kernel and recognize its shape without checking the parameter ranges and integrability.

For example, if $Y \mid \theta \sim N(\theta, 1)$ and $\pi(\theta) \propto 1$, then

$$\pi(\theta \mid y) \propto \exp\{-\tfrac{1}{2}(\theta - y)^2\},$$

which normalizes to $N(y, 1)$. The prior is improper but the posterior is proper. In contrast, for $X \mid \theta \sim U(0, \theta)$ on $\theta > 0$, the likelihood is $\theta^{-1}\mathbb{1}(\theta > x)$ for $x > 0$. With $\pi(\theta) \propto 1$, the posterior kernel $\theta^{-1}\mathbb{1}(\theta > x)$ has integral $\int_x^\infty \theta^{-1}\, \mathrm{d}\theta = \infty$, so the posterior is improper. A different prior proportional to $1/\theta$ gives a kernel $\theta^{-2}\mathbb{1}(\theta > x)$, which is integrable for $x > 0$.

5.3 Jeffreys' prior

Teacher's spoken explanation

Jeffreys' prior is constructed from the square root of Fisher information. Its key property is invariance under smooth one-to-one reparameterization. This makes it a systematic reference prior in many one-parameter problems, although it can be improper and it is not automatically the best prior for every scientific task.

For a scalar parameter θ with likelihood $f(y \mid \theta)$, the Fisher information in one observation is

$$\mathcal{I}(\theta) = \mathbb{E}_\theta\left[\left\{\frac{\partial}{\partial\theta}\log f(Y \mid \theta)\right\}^2\right] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\log f(Y \mid \theta)\right],$$

under regularity conditions. Jeffreys' prior is

$$\boxed{\pi_J(\theta) \propto \sqrt{\mathcal{I}(\theta)}.}$$

The square root is essential. For a vector parameter $\boldsymbol{\theta}$, the usual Jeffreys rule uses $\pi_J(\boldsymbol{\theta}) \propto \sqrt{\det \mathcal{I}(\boldsymbol{\theta})}$.

5.3.1 Why the square root of Fisher information?

Teacher's spoken explanation

Fisher information measures how sharply the likelihood can distinguish nearby parameter values. The square root appears because it transforms with the appropriate Jacobian factor under reparameterization. That is what gives Jeffreys' rule its invariance property.

If $\phi = g(\theta)$ is a smooth one-to-one reparameterization, then

$$\mathcal{I}_\phi(\phi) = \mathcal{I}_\theta(\theta) \left(\frac{\mathrm{d}\theta}{\mathrm{d}\phi} \right)^2.$$

Taking square roots gives

$$\sqrt{\mathcal{I}_\phi(\phi)}\,\mathrm{d}\phi = \sqrt{\mathcal{I}_\theta(\theta)} \left| \frac{\mathrm{d}\theta}{\mathrm{d}\phi} \right| \mathrm{d}\phi = \sqrt{\mathcal{I}_\theta(\theta)}\,\mathrm{d}\theta.$$

Thus the prior measure is invariant under the transformation. A constant prior is not generally invariant in this way.

5.3.2 Example 1: Bernoulli success probability

Teacher's spoken explanation

For a Bernoulli probability, calculate the Fisher information first and then take its square root. This produces a prior density proportional to one over the square root of theta times one minus theta. It is not uniform; it places more density near the boundaries while remaining integrable.

For one observation $Y \sim \text{Bernoulli}(\theta)$,

$$\ell(\theta; y) = y \log \theta + (1 - y) \log(1 - \theta),$$
$$\frac{\partial \ell}{\partial \theta} = \frac{y}{\theta} - \frac{1 - y}{1 - \theta}.$$

Its Fisher information is $\mathcal{I}(\theta) = 1/[\theta(1 - \theta)]$. Hence

$$\pi_J(\theta) \propto \frac{1}{\sqrt{\theta(1 - \theta)}},$$

which is the $\text{Beta}(1/2, 1/2)$ prior. After z successes in n trials, the posterior is $\text{Beta}(z+1/2, n-z+1/2)$.

5.3.3 Example 2: Poisson rate

Teacher's spoken explanation

For a Poisson rate, Fisher information is proportional to the reciprocal of the rate. Taking the square root gives a prior proportional to lambda to the power minus one half. We then check whether the resulting prior is proper; in this case it is not proper over the full positive half-line.

For $Y \sim \text{Poisson}(\lambda)$,

$$\ell(\lambda; y) = -\lambda + y \log \lambda - \log(y!), \quad \frac{\partial \ell}{\partial \lambda} = -1 + \frac{y}{\lambda}.$$

Since $\text{Var}(Y) = \lambda$, $\mathcal{I}(\lambda) = 1/\lambda$, so

$$\pi_J(\lambda) \propto \lambda^{-1/2}.$$

This is improper on $(0, \infty)$ without a further factor. Its posterior under a Poisson sample may nevertheless be proper and is $\text{Gamma}(\sum y_i + 1/2, n)$ for n unit exposures.

5.3.4 Example 3: Normal location with known variance

Teacher's spoken explanation

For a Normal location parameter with known variance, Fisher information is constant in the mean. Jeffreys' prior is therefore constant, which is an improper flat prior on the real line. The resulting posterior may nevertheless be proper when data are observed.

For $Y \sim N(\mu, \sigma^2)$ with known σ^2, the Fisher information for μ is $1/\sigma^2$, which is constant in μ. Jeffreys' prior is therefore $\pi_J(\mu) \propto 1$, a flat prior on the location parameter.

5.3.5 Example 4: Exponential rate

Teacher's spoken explanation

For an Exponential rate, compute the expected negative second derivative of the log likelihood or the variance of the score. The information is proportional to the reciprocal square of the rate, so the Jeffreys density is proportional to one over the rate. Again, inspect propriety on the parameter space.

For $Y \sim \text{Exponential}(\lambda)$ with rate parameter λ, one observation has log likelihood $\log \lambda - \lambda y$. The score is $1/\lambda - y$, and the Fisher information is $1/\lambda^2$. Thus $\pi_J(\lambda) \propto 1/\lambda$. Notice the difference between Poisson rate ($\lambda^{-1/2}$) and Exponential rate (λ^{-1}): Jeffreys' prior depends on the likelihood family, not just on the fact that both parameters are rates.

5.3.6 A caution for multi-parameter Normal models

Teacher's spoken explanation

In multi-parameter problems, different reference-prior constructions can lead to different answers depending on the parameter of interest and nuisance parameters. Do not transfer a one-parameter rule mechanically without checking the model and parameterization.

For $Y_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both μ and σ^2 unknown, different reference-prior constructions can lead to different joint prior kernels. The lecture notes explicitly use

$$\pi(\mu, \sigma^2) \propto \frac{1}{\sigma^2},$$

which yields the familiar posterior $\mu \mid \sigma^2, y \sim N(\overline{y}, \sigma^2/n)$ and $\sigma^2 \mid y \sim \text{Inv-}\chi^2(n-1, s^2)$, where $s^2 = \sum_i (y_i - \overline{y})^2/(n-1)$. This is often called a reference or independent-Jeffreys-type prior. The joint Jeffreys prior obtained from the determinant of the full two-parameter Fisher information is proportional to $(\sigma^2)^{-3/2}$. Do not silently interchange these two priors; follow the convention specified in the question or course.

5.4 Sensitivity to prior choice

Teacher's spoken explanation

Prior sensitivity asks whether conclusions change substantially under other reasonable priors. If the posterior is nearly unchanged, the data are influential relative to the prior. If the result changes materially, that dependence should be reported rather than hidden.

A prior can be proper but still strongly informative. For instance, Normal$(0, 10)$ and Normal$(0, 100)$ priors have the same center but different spreads. The narrower prior assigns relatively more density near zero and more strongly regularizes away from large coefficient values. Sensitivity analysis

compares posterior summaries or decisions under several defensible priors. This is especially important when the likelihood is weakly informative, the sample size is small, or the parameter is only weakly identified.

Teacher's spoken explanation

We have now made explicit the assumptions that enter before the data arrive. We know that a prior can simplify calculation, express genuine information, or serve as a reference construction; we also know that improper priors require special care, especially when a marginal likelihood is needed. For simple examples, the posterior can be derived exactly. In a realistic model with many dependent parameters, however, no amount of algebra may produce a convenient closed form. The next chapter addresses that computational barrier. We will keep the posterior as our target and learn how to explore it by simulation.

Chapter 6

Sampling from the posterior: MCMC, Gibbs and Metropolis–Hastings

Teacher's spoken explanation

Imagine that our model has become more realistic: several unknown parameters interact, and the posterior no longer belongs to a convenient named family. The Bayesian target has not changed. We still want the posterior distribution. What has changed is our ability to calculate it by hand. This is the point at which computation enters the story. Rather than abandon the posterior, we construct a Markov chain whose long-run behaviour allows us to approximate it. Gibbs sampling and Metropolis–Hastings are two ways to do this. But a computer returning a long list of numbers is not proof that the chain has explored the posterior properly. We must learn to read its graphical behaviour and diagnose convergence and mixing.

Teacher's spoken explanation

This chapter is about computational inference. We study Markov chains designed to have the posterior as their stationary distribution, learn the algorithms, and inspect their output before trusting it.

6.1 Why simulation is needed

Teacher's spoken explanation

In simple conjugate models, we can derive the posterior analytically. In more complicated models, the posterior may be known only up to a normalizing constant or may be too high-dimensional for direct integration. MCMC creates dependent draws whose long-run distribution is the posterior, allowing us to estimate posterior summaries numerically.

In a simple conjugate model, the posterior can be written exactly. In a complex model, the posterior normalizing constant or posterior integrals may be intractable. Markov chain Monte Carlo (MCMC) methods generate dependent draws whose long-run distribution is the target posterior. Posterior expectations can then be approximated by sample averages:

$$\mathbb{E}\{g(\theta) \mid y\} \approx \frac{1}{M} \sum_{m=1}^{M} g(\theta^{(m)}).$$

MCMC is not a new posterior distribution. It is a computational method for approximating summaries of the posterior already specified by the model and data.

6.2 Markov chains

Teacher's spoken explanation

A Markov chain is a stochastic sequence where the distribution of the next state depends on the present state, not the entire past, once the current state is known. In MCMC we design the transition rule so that the desired posterior is stationary. We then need to assess whether the chain has explored that target distribution adequately.

A sequence $\theta^{(0)}, \theta^{(1)}, \ldots$ is a Markov chain if

$$p(\theta^{(t+1)} \mid \theta^{(t)}, \theta^{(t-1)}, \ldots) = p(\theta^{(t+1)} \mid \theta^{(t)}).$$

The next state depends on the current state, not on the entire history. An MCMC transition rule is constructed so that the desired posterior $\pi(\theta \mid y)$ is a stationary distribution. Under appropriate irreducibility, aperiodicity and recurrence conditions, the chain converges to that target from suitable starting values.

6.3 Gibbs sampling

Teacher's spoken explanation

Gibbs sampling updates one unknown quantity at a time by drawing from its full conditional distribution given the current values of all other unknowns. Each update is exact for that conditional distribution. Repeating the updates produces a Markov chain with the joint posterior as its stationary distribution, under the usual regularity and irreducibility conditions.

Gibbs sampling is useful when the full conditional distributions are available in standard forms. Suppose the target posterior is $\pi(x, y \mid D)$. Its full conditionals are $\pi(x \mid y, D)$ and $\pi(y \mid x, D)$. One Gibbs sweep is:

1. Start from an initial value $x^{(0)}$ (and, if needed, $y^{(0)}$).
2. Draw $y^{(t+1)} \sim \pi(y \mid x^{(t)}, D)$.
3. Draw $x^{(t+1)} \sim \pi(x \mid y^{(t+1)}, D)$.
4. Repeat.

At each update, one block is sampled from its full conditional while other blocks are held at their latest values. The samples are dependent. A Gibbs sampler can be viewed as a special case of Metropolis–Hastings that accepts every proposed full-conditional draw.

6.3.1 A bivariate Normal conditional example

Teacher's spoken explanation

A multivariate Normal distribution has Normal conditional distributions. This makes it an ideal illustration of Gibbs sampling: draw the first coordinate conditional on the current second coordinate, then draw the second coordinate conditional on the newly updated first coordinate. A single sweep is one update of every coordinate.

Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2 \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right].$$

Then

$$X \mid Y = y \sim N(\rho y, 1-\rho^2), \qquad Y \mid X = x \sim N(\rho x, 1-\rho^2).$$

A Gibbs update alternates between these two Normal distributions. If $|\rho|$ is close to one, the conditional distributions are narrow relative to the elongated joint distribution; movement along the long direction can be slow, leading to high autocorrelation and poor mixing.

6.4 Metropolis–Hastings algorithm

Teacher's spoken explanation

Metropolis–Hastings proposes a new state from a proposal distribution and then accepts or rejects it using a probability that corrects for both the target density and asymmetry in the proposal. The correction is what ensures the desired stationary distribution. A rejected proposal means the chain stays at the current value; that repeated value is part of the chain, not a missing observation.

Metropolis–Hastings (MH) is more general: it needs the target posterior up to proportionality and a proposal distribution. From current state θ, propose $\theta^* \sim q(\theta^* \mid \theta)$. Accept the proposal with probability

$$\boxed{\alpha(\theta, \theta^*) = \min\left\{1, \frac{\pi(\theta^* \mid y) q(\theta \mid \theta^*)}{\pi(\theta \mid y) q(\theta^* \mid \theta)}\right\}.}$$

If accepted, set $\theta^{(t+1)} = \theta^*$; otherwise retain the current state, $\theta^{(t+1)} = \theta$. The unknown posterior normalizing constant cancels from the ratio, so an unnormalized posterior kernel is sufficient.

6.4.1 Symmetric random-walk proposal

Teacher's spoken explanation

With a symmetric random-walk proposal, the proposal density cancels from the acceptance ratio. The acceptance probability becomes the smaller of one and the ratio of target densities at the proposed and current states. Symmetry simplifies the formula, but it does not remove the need to check whether the chain mixes well.

If $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$, the proposal terms cancel:

$$\alpha(\theta, \theta^*) = \min\left\{1, \frac{\pi(\theta^* \mid y)}{\pi(\theta \mid y)}\right\}.$$

For a random-walk proposal, one often uses $\theta^* = \theta + \varepsilon$ with ε drawn from a centered symmetric distribution. If a proposal falls outside the parameter space, its target density is zero and it is rejected.

6.4.2 Why the proposal correction matters

Teacher's spoken explanation

If the proposal is asymmetric, moving from the current state to the proposed state may not be as likely as the reverse move. The Hastings correction accounts for that imbalance. Omitting it generally changes the stationary distribution and can make the algorithm target the wrong posterior.

For an asymmetric proposal, the reverse proposal probability need not equal the forward probability. Ignoring the ratio $q(\theta \mid \theta^*)/q(\theta^* \mid \theta)$ changes the stationary distribution and generally produces the

wrong chain. The correction enforces detailed balance:

$$\pi(\theta)q(\theta^* \mid \theta)\alpha(\theta, \theta^*) = \pi(\theta^*)q(\theta \mid \theta^*)\alpha(\theta^*, \theta).$$

6.5 Convergence and mixing diagnostics: graphical emphasis

Teacher's spoken explanation

We never declare convergence merely because the algorithm has run for a large number of iterations. We inspect several graphical summaries. The trace plot checks stability over iteration, the density plot checks the sampled shape, the autocorrelation plot checks dependence between successive draws, and multiple chains help reveal whether starting values lead to different regions.
A reliable conclusion uses several diagnostics together. No single plot or numerical threshold proves convergence in every possible model.

MCMC output is not trustworthy merely because a large number of iterations were run. A chain may remain near its starting value, explore only one mode, or move so slowly that its effective information is small. Graphical summaries are essential.

6.5.1 Trace plot

Teacher's spoken explanation

A trace plot places the sampled parameter value against iteration. A healthy trace usually moves around a stable band, with no persistent drift and no long periods stuck at one value. A trend, a sudden shift or long flat sections are warning signs.

Plot iteration number on the horizontal axis and sampled parameter values on the vertical axis. A healthy trace should look like a stable, well-mixed horizontal band after the initial transient. Warning signs include a persistent trend, long flat stretches, sudden jumps between separated regions with long gaps, or different chains occupying different regions.

6.5.2 Posterior density or histogram

Teacher's spoken explanation

A histogram or density estimate summarizes the distribution of the retained draws. We compare the shapes across chains. Strongly different shapes may indicate that chains have not explored the same posterior region.

Plot the marginal posterior sample distribution. Overlay density estimates from several chains. Similar shapes support the view that chains have explored the same marginal distribution. Different modes, centers, spreads or tails indicate incomplete mixing or non-convergence.

6.5.3 Autocorrelation function (ACF)

Teacher's spoken explanation

Autocorrelation measures how strongly a draw is associated with draws at later lags. Slowly decaying autocorrelation means that nearby iterations carry similar information. This reduces effective sample size, even if the raw iteration count is large.

The lag-k autocorrelation is the correlation between draws k iterations apart. High positive autocorrelation over many lags means successive draws contain substantial repeated information. A slowly decaying ACF is a warning of poor mixing.

6.5.4 Running mean and multiple chains

Teacher's spoken explanation

The running mean shows whether an estimate stabilizes as more iterations are included. Multiple chains started from dispersed initial values should eventually explore similar regions and produce similar summaries. A stable running mean alone is not enough, but it is a useful visual diagnostic.

A running mean plot checks whether estimated posterior summaries stabilize as iterations accumulate. Multiple chains should start from dispersed initial values. Their trace plots and marginal distributions should overlap and explore the same regions. Agreement across chains is useful but not a proof of convergence.

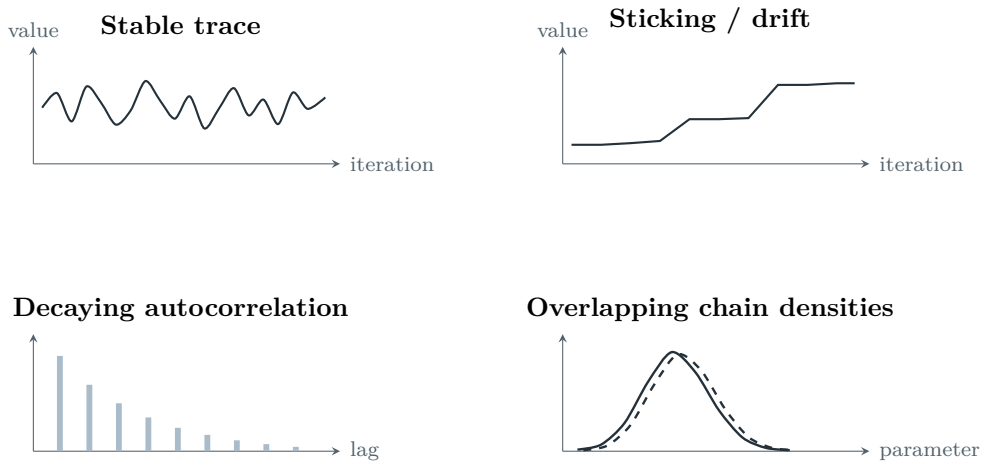


Figure 6.1: Schematic graphical diagnostics. A stable trace, decaying autocorrelation and overlapping chain densities are reassuring. Long flat segments, trends or separated chain distributions indicate poor mixing or possible non-convergence. The curves are illustrative, not output from a fitted model.

6.5.5 Effective sample size and Monte Carlo standard error

Teacher's spoken explanation

Effective sample size translates autocorrelated draws into the approximate number of independent draws with comparable information. Monte Carlo standard error then quantifies numerical uncertainty in an estimated posterior summary. We should report these quantities when simulation accuracy matters.

For M correlated draws with lag autocorrelations ρ_k, a common approximation is

$$\mathrm{ESS} \approx \frac{M}{1 + 2\sum_{k=1}^{\infty} \rho_k}.$$

For a sample mean with marginal posterior standard deviation s, the Monte Carlo standard error is approximately $s/\sqrt{\mathrm{ESS}}$. Thus $M = 10,000$ stored draws can contain far less information than 10,000 independent draws if autocorrelation is high.

6.5.6 Potential scale reduction factor

Teacher's spoken explanation

The potential scale reduction statistic compares variation within chains with variation between chains. Values close to one are reassuring when chains also look stable and effective sample sizes are adequate. It is a diagnostic rather than a proof, so never quote it without graphical inspection.

The Gelman–Rubin family of diagnostics compares within-chain and between-chain variation. Split-$\widehat{R}$ near one indicates that chains have similar scale after splitting, but it must be combined with trace plots, ESS, ACF and adequate exploration. A value near one is a diagnostic, not a mathematical guarantee.

Exam check

A graphical-diagnostics answer should name the plot, say what a good plot looks like, and say what a warning pattern means. Example: "In the trace plot, a stable band with frequent movement and no long trend suggests stationarity and mixing; a sustained drift or long flat segments suggests non-convergence or poor mixing."

6.6 Burn-in, thinning and practical cautions

Teacher's spoken explanation

Burn-in removes an initial part of the chain that may reflect the starting value. Thinning keeps only selected draws. Thinning is not a remedy for non-convergence and is often unnecessary for estimating posterior summaries. The priority is to achieve and diagnose good exploration, not merely to produce a smaller file of draws.

The initial portion of a chain may depend strongly on the starting value; discarding an initial burn-in is a common practical step. Thinning keeps only every kth draw, but it does not repair a chain that has not converged and often wastes samples; it is not automatically needed for posterior estimation. Good practice is to run multiple chains, inspect all diagnostics, report effective sample sizes and Monte Carlo errors, and make sure the chain moves between all relevant posterior regions.

6.7 Posterior predictive checks

Teacher's spoken explanation

Posterior predictive checking asks whether data simulated from the fitted model resemble the observed data in scientifically important ways. We draw parameters from the posterior, simulate replicated data, and compare summaries such as means, variances, extremes or group differences. Poor replication indicates a modelling weakness, not a failure of Bayes' theorem.

After drawing $\theta^{(m)}$ from the posterior, simulate replicated data $Y^{rep,(m)} \sim p(y \mid \theta^{(m)})$. Compare a statistic $T(Y)$ to $T(Y^{rep})$ across replications. Examples include sample mean, variance, maximum, number of zeros, tail counts, group differences or residual patterns. A systematic discrepancy indicates that the likelihood or hierarchical structure may not reproduce an important feature of the observed data.

Teacher's spoken explanation

MCMC has given us a practical route to posterior summaries when direct integration is difficult. We have learned that convergence is not established by a large iteration count alone: trace plots, density overlays, autocorrelation, running means, effective sample size and agreement across chains all matter. We are now ready to use Bayesian inference for a more structured scientific task. Rather than estimating one parameter such as a single recovery probability, we will relate an outcome to several predictors. This takes us to Bayesian linear models, where priors also become a way to control unstable estimation and express uncertainty about model structure.

Chapter 7

Bayesian linear models, regularization and variable selection

Teacher's spoken explanation

We can now represent uncertainty even when an analytic posterior is difficult. Let us return to the hospitals and ask a richer scientific question: why do patient outcomes differ? A single probability or mean cannot explain the roles of age, baseline condition, treatment and other predictors. Linear models give us a structured way to relate an outcome to several covariates. The Bayesian contribution is to place distributions on regression coefficients and carry uncertainty through the fit. This also reveals a deep connection: a Gaussian prior produces a ridge-type penalty, while a Laplace prior produces the LASSO penalty at the MAP level. When many predictors compete, spike-and-slab priors let us express uncertainty about whether a variable belongs in the model at all.

Teacher's spoken explanation

We extend Bayesian reasoning to regression. A vector of coefficients replaces a single parameter, and matrix algebra makes the structure compact. We connect Normal priors to ridge, Laplace priors to the LASSO mode, and spike-and-slab priors to variable selection.

7.1 The Gaussian linear model

Teacher's spoken explanation

We now move from one unknown parameter to a vector of regression coefficients. The response vector is modelled as a linear mean X beta plus Gaussian error. The design matrix describes the predictors, beta contains the unknown effects, and sigma squared measures residual variability. Bayesian regression places a prior on these unknown quantities and updates it with the observed response.

The linear regression model in matrix form is

$$\boxed{Y = X\beta + \varepsilon, \qquad \varepsilon \sim N_n(0, \sigma^2 I_n).}$$

Here Y is an $n \times 1$ response vector, X is an $n \times p$ design matrix, β is a $p \times 1$ coefficient vector, and σ^2 is the error variance. Conditional on X, β, σ^2,

$$Y \mid \beta, \sigma^2, X \sim N_n(X\beta, \sigma^2 I_n),$$

with likelihood kernel

$$L(\beta, \sigma^2 \mid y) \propto (\sigma^2)^{-n/2} \exp\left\{-\frac{(y - X\beta)^\mathsf{T}(y - X\beta)}{2\sigma^2}\right\}.$$

Bayesian inference needs a prior for the regression coefficients and, if not fixed, a prior for the error variance. A common simplification is to use independent or conditionally independent priors for these unknown quantities.

7.2 Normal prior on coefficients: exact posterior when variance is known

Teacher's spoken explanation

A multivariate Normal prior on the coefficient vector is conjugate to a Gaussian likelihood with known variance. The posterior is another multivariate Normal distribution. Its precision matrix is the sum of prior precision and likelihood precision, and its mean balances prior information with the least-squares information in the data.

Assume σ^2 is known and

$$\beta \sim N_p(\beta_0, V_0),$$

where V_0 is positive definite. Combining the Normal likelihood and Normal prior gives

$$V_n = (V_0^{-1} + X^\mathsf{T}X/\sigma^2)^{-1}, \qquad \beta_n = V_n(V_0^{-1}\beta_0 + X^\mathsf{T}y/\sigma^2),$$

$$\boxed{\beta \mid y, X, \sigma^2 \sim N_p(\beta_n, V_n).}$$

The posterior mean and posterior mode coincide for this multivariate Normal posterior. If the prior is $N_p(0, \tau^2 I_p)$, then

$$V_n = (X^\mathsf{T}X/\sigma^2 + I_p/\tau^2)^{-1}, \qquad \beta_n = (X^\mathsf{T}X + \sigma^2\tau^{-2}I_p)^{-1}X^\mathsf{T}y.$$

This is the ridge estimator with penalty parameter $\lambda = \sigma^2/\tau^2$.

7.3 Ridge regression as a Bayesian MAP estimator

Teacher's spoken explanation

Ridge regression minimizes residual sum of squares plus a squared-coefficient penalty. A zero-centred Normal prior on coefficients contributes a quadratic term to the negative log posterior. Maximizing the posterior is therefore equivalent to ridge minimization, with the prior variance controlling the penalty strength. This gives a probabilistic interpretation of shrinkage.

The Gaussian prior is

$$\beta_j \overset{\text{iid}}{\sim} N(0, \tau^2).$$

The negative log posterior, ignoring constants, is

$$\frac{1}{2\sigma^2}\|y - X\beta\|^2 + \frac{1}{2\tau^2}\|\beta\|^2.$$

Multiplying by $2\sigma^2$ shows that the MAP estimate minimizes

$$\boxed{\|y - X\beta\|^2 + \lambda\|\beta\|^2, \qquad \lambda = \frac{\sigma^2}{\tau^2}.}$$

A narrower prior (smaller τ^2) gives a larger penalty and stronger shrinkage toward zero. As $\tau^2 \to \infty$, the prior becomes flat in the coefficient directions and the MAP approaches least squares when the design has full column rank. As $\tau^2 \to 0$, the coefficients are concentrated near zero. In most regression implementations, the intercept is not penalized; this corresponds to assigning it a diffuse prior rather than the same shrinkage prior as the slopes.

7.4 LASSO as a Bayesian MAP estimator

Teacher's spoken explanation

The LASSO penalty is the sum of absolute coefficient values. A Laplace prior has a density whose negative log is proportional to the absolute value, so the posterior mode solves the LASSO optimization problem. The prior favours coefficients near zero and can produce exact zeros at the mode, although a continuous posterior does not assign positive probability to a coefficient being exactly zero.

For known σ^2, assign independent Laplace priors to the coefficients:

$$\pi(\beta_j \mid \sigma^2, \lambda) = \frac{\lambda}{2\sigma^2} \exp\left(-\frac{\lambda|\beta_j|}{\sigma^2}\right).$$

The joint prior kernel is $\exp\{-\lambda \|\beta\|_1 / \sigma^2\}$. The negative log posterior is proportional to

$$\frac{1}{2\sigma^2} \|y - X\beta\|^2 + \frac{\lambda}{\sigma^2} \|\beta\|_1 .$$

Therefore the MAP estimate minimizes

$$\boxed{\|y - X\beta\|^2 + 2\lambda \|\beta\|_1 .}$$

The penalty encourages some coefficients to be exactly zero because the L_1 penalty has a non-differentiable corner at zero. The scale parameter of the Laplace prior determines the penalty strength. If a different Laplace scale is used, the numerical relationship between the prior parameter and the LASSO tuning constant changes; derive the mapping from the stated prior rather than memorizing a symbol.

7.4.1 Bayesian LASSO as a Normal–Exponential mixture

Teacher's spoken explanation

The Laplace distribution can be represented as a scale mixture of Normals with an Exponential mixing distribution. This representation introduces latent local variances and makes conditional distributions convenient for Gibbs sampling. It explains why a model that looks non-Gaussian marginally can still have tractable conditional updates.

A Laplace distribution can be represented as a scale mixture of Normals. One common hierarchy is

$$\beta_j \mid \tau_j^2, \sigma^2 \sim N(0, \sigma^2\tau_j^2), \qquad \tau_j^2 \mid \lambda^2 \sim \text{Exponential}(\lambda^2/2).$$

Integrating out τ_j^2 gives a Laplace marginal prior for β_j (with rate λ/σ under this convention). This representation is useful because the conditional distribution of β is multivariate Normal, while local scale variables can be updated with Gibbs or Metropolis–Hastings steps. The local scales permit different degrees of shrinkage for different coefficients.

7.5 Elastic-net connection

Teacher's spoken explanation

Elastic net combines an absolute-value penalty and a squared penalty. In Bayesian language, this corresponds to combining the shrinkage behaviours associated with Laplace and Normal priors. The point of the connection is interpretive: penalties can be understood as prior assumptions about the size and sparsity of coefficients.

Combining a Laplace prior and a Gaussian prior gives a posterior mode corresponding to an elastic-net objective of the form

$$\|y - X\beta\|^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 .$$

The L_1 component encourages sparsity; the L_2 component stabilizes correlated predictors and shrinks coefficients continuously. The two penalty constants are determined by the exact prior parameterization and likelihood scale.

7.6 Normal–Inverse-Gamma prior for Bayesian regression

Teacher's spoken explanation

The Normal–Inverse-Gamma prior jointly models the regression coefficients and residual variance. Conditional on the variance, coefficients have a multivariate Normal prior; the variance has an inverse-Gamma prior. Conjugacy lets us update the coefficient covariance, posterior centre and variance hyperparameters using familiar matrix quantities.

Let

$$\beta \mid \sigma^2 \sim N_p(\beta_0, \sigma^2 V_0), \qquad \sigma^2 \sim \text{InvGamma}(a_0, b_0).$$

This is the conjugate Normal–Inverse-Gamma prior for the Gaussian linear model. Define

$$V_n = (V_0^{-1} + X^\mathsf{T} X)^{-1}, \qquad \beta_n = V_n(V_0^{-1}\beta_0 + X^\mathsf{T} y),$$

$$a_n = a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{1}{2}\left(y^\mathsf{T} y + \beta_0^\mathsf{T} V_0^{-1}\beta_0 - \beta_n^\mathsf{T} V_n^{-1}\beta_n\right).$$

Then

$$\boxed{\beta \mid \sigma^2, y \sim N_p(\beta_n, \sigma^2 V_n), \qquad \sigma^2 \mid y \sim \text{InvGamma}(a_n, b_n).}$$

The expression for b_n can also be written as prior scale plus half the posterior residual sum of squares after completing the square. The expression is nonnegative under a valid prior and design. For the special prior $\beta_0 = 0, V_0 = I_p$, the lecture notes' formulas reduce to $V_n = (X^\mathsf{T} X + I_p)^{-1}$ and $\beta_n = V_n X^\mathsf{T} y$.

7.6.1 Posterior prediction in Bayesian regression

Teacher's spoken explanation

For a new predictor vector, the predicted response depends on the uncertain coefficient vector and on future observation noise. Integrating over the posterior produces a predictive distribution. We must distinguish uncertainty about the fitted mean from uncertainty about a new individual response; the latter includes residual noise and is wider.

For a new design row x_*^T, the predictive distribution under the NIG model is

$$\widetilde{Y}_* \mid y, X, x_* \sim t_{2a_n}\left(x_*^\mathsf{T}\beta_n, \sqrt{\frac{b_n}{a_n}\left(1 + x_*^\mathsf{T} V_n x_*\right)}\right).$$

The 1 inside the parentheses represents future observation noise; $x_*^\mathsf{T} V_n x_*$ represents coefficient uncertainty. If the question asks for uncertainty in the latent mean $x_*^\mathsf{T}\beta$ rather than a future observed response, omit the 1 term.

7.7 The common improper regression prior and its conditions

Teacher's spoken explanation

A common reference prior is flat in the coefficients and proportional to one over sigma squared for the variance. It can yield a proper posterior only when the design matrix has full column rank and enough residual degrees of freedom remain. Always state these conditions; an algebraic formula is not enough if the posterior fails to normalize.

The lecture notes examine the prior kernel $\pi(\beta, \sigma^2) \propto 1/\sigma^2$. Under the Gaussian model, if X has full column rank p, $n > p$ and the least-squares residual sum of squares $\mathrm{SSE} > 0$, the posterior is proper and

$$\beta \mid \sigma^2, y \sim N_p\left(\widehat{\beta}, \sigma^2 (X^\mathsf{T} X)^{-1}\right),$$

$$\sigma^2 \mid y \sim \mathrm{InvGamma}\left(\frac{n-p}{2}, \frac{\mathrm{SSE}}{2}\right), \quad \widehat{\beta} = (X^\mathsf{T} X)^{-1} X^\mathsf{T} y.$$

The marginal posterior of β is multivariate Student-t with $n - p$ degrees of freedom, location $\widehat{\beta}$, and scale matrix $[\mathrm{SSE}/(n-p)](X^\mathsf{T} X)^{-1}$. If $\mathrm{rank}(X) < p$, the likelihood does not identify all coefficient directions; integration over a null direction is infinite, so this improper prior does not produce a proper posterior. The requirements $n > p$ and $\mathrm{SSE} > 0$ are also needed for the variance integral to be finite under this formulation.

7.8 Spike-and-slab prior and variable selection

Teacher's spoken explanation

Spike-and-slab puts a mixture prior on each coefficient: a spike concentrated at zero represents exclusion, and a broader slab represents inclusion with a nonzero effect. The model therefore has an explicit indicator for whether a predictor is active. This differs from ordinary shrinkage, which pulls coefficients toward zero without necessarily assigning exact posterior mass to zero.

Suppose many predictors are available, but only some are truly associated with the response. Let $\gamma_j \in \{0, 1\}$ indicate whether predictor j is included. A spike-and-slab prior is

$$\gamma_j \sim \mathrm{Bernoulli}(\pi), \qquad \beta_j \mid \gamma_j = \begin{cases} 0, & \gamma_j = 0, \\ N(0, \tau_{\mathrm{slab}}^2), & \gamma_j = 1. \end{cases}$$

The spike is a point mass at zero; the slab is a relatively diffuse distribution for a nonzero coefficient. A continuous approximation uses a narrow Normal spike $N(0, \tau_{\mathrm{spike}}^2)$ with small variance and a wider Normal slab.

7.8.1 Posterior inclusion probability

Teacher's spoken explanation

The posterior inclusion probability is the posterior probability that a variable's indicator equals one. It summarizes evidence for including the predictor while accounting for uncertainty in the coefficient and other model components. A high inclusion probability indicates strong posterior support under the chosen prior and likelihood.

The quantity

$$\mathbb{P}(\gamma_j = 1 \mid y)$$

is the posterior probability that predictor j is included. It combines the prior inclusion probability and the marginal likelihood under inclusion/exclusion:

$$\frac{\mathbb{P}(\gamma_j = 1 \mid y)}{\mathbb{P}(\gamma_j = 0 \mid y)} = \frac{p(y \mid \gamma_j = 1)}{p(y \mid \gamma_j = 0)} \times \frac{\pi}{1-\pi}.$$

This is a Bayes-factor update for the inclusion indicator. Variable selection can be based on a threshold, posterior model probabilities, or a decision rule that accounts for the costs of false inclusion and false exclusion. A coefficient's posterior mean being small is not the same as an explicit posterior probability of exclusion.

7.8.2 Gibbs idea for a spike-and-slab regression

Teacher's spoken explanation

A Gibbs sampler alternates between updating coefficient values given inclusion indicators and updating indicators given coefficient values and the data. The exact conditional formulas depend on the chosen slab and model specification. The conceptual point is that discrete model-selection uncertainty can be sampled jointly with continuous regression coefficients.

A Gibbs-type algorithm alternates among coefficient blocks, inclusion indicators, variance parameters and possibly local scales. When updating γ_j, condition on the current values of the other coefficients and compare the conditional posterior probabilities of $\gamma_j = 0$ and $\gamma_j = 1$. When updating β_j, it is exactly zero under the spike and drawn from its conditional continuous distribution under the slab. The lecture notes' coordinate-wise residual formulation is an efficient way to update one coefficient at a time: remove the current contribution $x_j\beta_j$ from the fitted value, form the partial residual, and compute the two conditional model probabilities.

Exam check

Ridge, LASSO and spike-and-slab all shrink coefficients, but their mechanisms differ. Ridge uses a Gaussian prior and continuous shrinkage; LASSO uses a Laplace prior and a MAP objective with an L_1 penalty; spike-and-slab uses a mixture with a point-mass spike and directly models inclusion/exclusion.

Teacher's spoken explanation

We have moved from one unknown parameter to a vector of regression coefficients, and from estimating all coefficients equally to controlling complexity and even questioning whether a predictor should be included. The Gaussian prior and Laplace prior show that familiar regularization methods can be understood as Bayesian MAP procedures, while spike-and-slab priors make inclusion itself uncertain. Yet our hospital investigation still has one unresolved feature: the patients are not independent islands. They are nested in groups that share conditions. A regression model with one common coefficient vector may not adequately describe that structure.

The final theory chapter introduces hierarchical models to represent group-level variation and controlled sharing of information.

Chapter 8

Hierarchical Bayes, empirical Bayes and profile likelihood

Teacher's spoken explanation

There is one more feature of our hospital story that we have not yet used. Patients are grouped within wards, and wards are grouped within hospitals. Two patients in the same hospital may share staff, protocols and local conditions. If we pretend that every group is unrelated, we discard information. If we force all groups to be identical, we discard real variation. Hierarchical modelling offers a middle path: groups have their own parameters, but those parameters are connected through a common population distribution. This produces partial pooling and shrinkage. We will then compare full hierarchical Bayes, which propagates uncertainty in the population-level parameters, with empirical Bayes, which estimates them and plugs them in. Profile likelihood provides a related likelihood-based way to handle nuisance parameters.

Teacher's spoken explanation

We study data organized into groups and ask how to learn about each group while allowing groups to share information. We compare pooling strategies, derive shrinkage, and distinguish full hierarchical Bayes from empirical Bayes and profile likelihood.

8.1 What is a hierarchical model?

Teacher's spoken explanation

A hierarchical model introduces parameters at more than one level. Observations within a group depend on a group-specific parameter, and the group parameters share a population distribution governed by hyperparameters. This structure allows information to be shared across groups while preserving group-level differences.

A hierarchical model introduces parameters at more than one level. A generic structure is

$$Y_{ij} \mid \theta_j \sim f(\cdot \mid \theta_j), \qquad \theta_j \mid \phi \sim g(\cdot \mid \phi), \qquad \phi \sim h(\phi).$$

Here observations Y_{ij} belong to group j, group-specific parameters are θ_j, and hyperparameters ϕ describe the distribution of the group parameters. The joint distribution factors as

$$p(\phi, \theta_1, \ldots, \theta_J, y) = p(\phi) \prod_{j=1}^{J} p(\theta_j \mid \phi) \prod_{i=1}^{n_j} p(y_{ij} \mid \theta_j).$$

The posterior is obtained by multiplying this joint distribution and conditioning on all observed data.

8.2 Complete pooling, no pooling and partial pooling

Teacher's spoken explanation

Complete pooling treats all groups as if they share one common parameter. No pooling estimates each group independently. Partial pooling lies between these extremes: group estimates are allowed to differ, but they are informed by a common population distribution. This is the central practical advantage of hierarchical modelling.

- **Complete pooling:** treat all groups as having one common parameter. Differences between groups are ignored.
- **No pooling:** estimate each group separately, with no information shared across groups.
- **Partial pooling:** give group parameters a common distribution. Each group estimate uses its own data and is shrunk toward the population-level distribution.

Partial pooling is a central reason for hierarchical modeling. Small groups are shrunk more strongly toward the overall mean; large groups are allowed to be driven more by their own data.

8.3 Normal–Normal hierarchical model and shrinkage

Teacher's spoken explanation

In the Normal–Normal hierarchy, each group estimate has a Normal likelihood and its group mean has a Normal prior centred at a population mean. The posterior mean becomes a weighted average of the group estimate and the population mean. Small or noisy groups are shrunk more strongly toward the population centre; precise groups are allowed to remain closer to their own data.

Suppose the group sample mean satisfies

$$\overline{Y}_j \mid \theta_j \sim N\left(\theta_j, \frac{\sigma^2}{n_j}\right), \qquad \theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2),$$

where σ^2, μ, τ^2 are temporarily treated as known. Completing the square gives

$$\theta_j \mid \overline{y}_j, \mu, \tau^2 \sim N(m_j, v_j),$$

$$v_j = \left(\frac{n_j}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}, \qquad m_j = v_j\left(\frac{n_j \overline{y}_j}{\sigma^2} + \frac{\mu}{\tau^2}\right).$$

Equivalently,

$$m_j = w_j \overline{y}_j + (1 - w_j)\mu, \qquad w_j = \frac{n_j \tau^2}{n_j \tau^2 + \sigma^2}.$$

When n_j is large, w_j is near one and the group estimate is close to its sample mean. When n_j is small or within-group noise σ^2 is large, the estimate is pulled more strongly toward μ. This is partial pooling.

8.4 Hierarchical Bayes

Teacher's spoken explanation

In full hierarchical Bayes, we place a prior on the hyperparameters as well as a prior on group-level parameters conditional on those hyperparameters. We then infer all unknown levels jointly. Uncertainty about the population distribution is carried through to the group-level posterior rather than replaced by a single plug-in estimate.

In fully Bayesian hierarchical inference, a hyperprior is assigned to the hyperparameters, for example

$$\mu \sim p(\mu), \qquad \tau^2 \sim p(\tau^2),$$

and inference is based on the joint posterior of all group parameters and hyperparameters:

$$p(\theta_1, \ldots, \theta_J, \mu, \tau^2 \mid y).$$

Uncertainty about μ and τ^2 is propagated into the posterior of each θ_j. Posterior summaries integrate over hyperparameter uncertainty rather than treating estimated hyperparameters as fixed.

8.5 Empirical Bayes

Teacher's spoken explanation

Empirical Bayes estimates hyperparameters from the observed data, often by maximizing the marginal likelihood after integrating out the group-level parameters. We then use those estimated hyperparameters in the posterior for each group. This can be computationally convenient, but it usually treats estimated hyperparameters as fixed and so may understate uncertainty compared with full hierarchical Bayes.

Empirical Bayes estimates some or all hyperparameters from the observed data and then plugs those estimates into the prior. For example, the marginal likelihood of the group data is

$$m(y \mid \mu, \tau^2) = \prod_{j=1}^{J} \int p(y_j \mid \theta_j) p(\theta_j \mid \mu, \tau^2) \, \mathrm{d}\theta_j.$$

An empirical Bayes estimate is

$$(\widehat{\mu}, \widehat{\tau}^2) = \arg\max_{\mu, \tau^2} m(y \mid \mu, \tau^2).$$

The group posterior is then computed using $\widehat{\mu}, \widehat{\tau}^2$ as fixed values. Empirical Bayes is computationally convenient and shares information across groups, but plug-in inference does not fully propagate uncertainty in the estimated hyperparameters. Fully Bayesian inference instead places a hyperprior on these quantities and integrates over their posterior.

8.6 Profile likelihood

Teacher's spoken explanation

Profile likelihood handles nuisance parameters by maximizing the likelihood over them for each fixed value of the parameter of interest. It is not the same as integrating the nuisance parameter out. Maximization and integration answer different questions, so profile likelihood should not be

confused with Bayesian marginal likelihood.

Suppose a likelihood depends on a parameter of interest ψ and nuisance parameter λ. The **profile likelihood** is

$$\boxed{L_p(\psi) = \max_{\lambda} L(\psi, \lambda).}$$

For each fixed ψ, maximize the likelihood over the nuisance parameter. This produces a function of ψ alone. It is not the same as Bayesian marginalization:

$$\text{profile likelihood: } \max_{\lambda} L(\psi, \lambda), \qquad \text{Bayesian marginal likelihood: } \int L(\psi, \lambda)\pi(\lambda \mid \psi)\,\mathrm{d}\lambda.$$

Maximization selects the best-fitting nuisance value; integration averages over nuisance uncertainty using a prior. Do not confuse profile likelihood, marginal likelihood and posterior marginal distribution.

8.7 Random-intercept model as a hierarchical example

Teacher's spoken explanation

A random-intercept model gives each group its own intercept, while assuming these intercepts come from a common population distribution. It is a concrete example of partial pooling. Groups with little information are pulled toward the population mean, while groups with more information are estimated more individually.

The lecture notes introduce a model of the form

$$Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_i + \varepsilon_{ij},$$

where i indexes individuals or groups and j indexes repeated observations within an individual. A standard specification is

$$b_i \overset{\text{iid}}{\sim} N(0, \tau^2), \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

with random effects independent of within-individual errors. The population intercept and slope describe the overall relationship; b_i allows individual-specific vertical shifts. Observations from the same individual are correlated because they share b_i:

$$\mathrm{Cov}(Y_{ij}, Y_{ik}) = \tau^2 \quad (j \neq k),$$

while observations from different individuals have covariance zero under the stated independence structure. The intraclass correlation is

$$\mathrm{ICC} = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

A hierarchical prior on random effects yields partial pooling: an individual's estimated intercept is informed by both that individual's repeated observations and the population distribution of effects.

Teacher's spoken explanation

We have reached the most layered view of the investigation. A patient-level outcome is described using a model; groups have their own parameters; and those group parameters are themselves linked through population-level quantities. Partial pooling follows from this structure rather than being an arbitrary compromise. Full hierarchical Bayes carries uncertainty at every level, whereas empirical Bayes estimates the hyperparameters before updating the groups. Profile likelihood offers another way to eliminate nuisance parameters when constructing a likelihood for the parameter of interest. With this final layer, the course has travelled from a single unknown

probability to a multi-level statistical model. We will now turn that conceptual journey into a disciplined way of answering examination questions.

Chapter 9

Exam method, formula sheet and common mistakes

Teacher's spoken explanation

We have now reached the point where the individual ideas form a single toolkit. The examination may present a coin, a waiting-time study, a regression problem or a collection of groups. Do not begin by searching your memory for a formula. Begin by identifying the data-generating model and the unknown quantity. Then identify the likelihood, the prior and the target of the question. Are we updating, estimating, predicting, comparing hypotheses, simulating, selecting variables or modelling groups? Once that decision is made, the mathematical route becomes much clearer. This chapter turns the story into a repeatable examination method.

Teacher's spoken explanation

This chapter turns the theory into a repeatable examination method. Every solution should make the model, prior, likelihood, posterior and requested summary transparent.

9.1 A reliable derivation template

Teacher's spoken explanation

In an examination, the cleanest solution is one where the marker can see every modelling decision. Begin with the sampling distribution, write the likelihood, state the prior with its parameterization, multiply the kernels, identify the posterior family, report updated parameters, and then calculate the requested quantity. This sequence prevents unexplained jumps.

For almost every conjugate update, use this template:

1. State the sampling distribution and parameterization, e.g. $Y_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$.
2. Write the prior including its support and hyperparameters.
3. Write the likelihood as a function of the unknown parameter, dropping only factors independent of that parameter.
4. Multiply prior and likelihood.
5. Collect powers and exponential terms.
6. Match the kernel to a named distribution.
7. State the posterior distribution and every updated hyperparameter.

8. If asked, calculate posterior summaries, a credible interval, a Bayes factor or a predictive distribution using the posterior.

9.2 Compact formula sheet

Teacher's spoken explanation

This formula section is a map, not a replacement for understanding. When a formula is used, say which model assumptions make it valid and define every parameter. In particular, check whether a Gamma second parameter is a rate or a scale, and whether a Normal prior is written in terms of variance or precision.

Core formulas

Bayes rule: $p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$.
Posterior odds: Posterior odds $= \mathrm{BF}_{10} \times$ Prior odds.
Beta–Bernoulli: $\mathrm{Beta}(\alpha, \beta) \to \mathrm{Beta}(\alpha + z, \beta + n - z)$.
Gamma–Poisson: $\mathrm{Gamma}(a, b) \to \mathrm{Gamma}(a + \sum y_i, b + n)$ for unit exposures.
Gamma–Exponential: $\mathrm{Gamma}(a, b) \to \mathrm{Gamma}(a + n, b + \sum y_i)$ for rate parameter.
Dirichlet–Multinomial: $\mathrm{Dirichlet}(\boldsymbol{\alpha}) \to \mathrm{Dirichlet}(\boldsymbol{\alpha} + \boldsymbol{n})$.
Normal–Normal precision: $\tau_n^{-2} = \tau_0^{-2} + n\sigma^{-2}$ and $\mu_n = \tau_n^2(\mu_0/\tau_0^2 + n\overline{y}/\sigma^2)$.
Jeffreys: $\pi_J(\theta) \propto \sqrt{\mathcal{I}(\theta)}$.
MH acceptance: $\alpha = \min\{1, \pi(\theta^*)q(\theta \mid \theta^*)/[\pi(\theta)q(\theta^* \mid \theta)]\}$.
Ridge MAP: $\arg\min_\beta\{\|y - X\beta\|^2 + \lambda \|\beta\|_2^2\}$.
LASSO MAP: $\arg\min_\beta\{\|y - X\beta\|^2 + \lambda \|\beta\|_1\}$ under the corresponding Laplace-prior scale.
Profile likelihood: $L_p(\psi) = \max_\lambda L(\psi, \lambda)$.

9.3 Common mistakes to avoid

Teacher's spoken explanation

Let us finish the review of methods by naming the errors that are easiest to make under examination pressure: reversing conditional probabilities, dropping exposure, mixing Gamma rate and scale, confusing variance and precision, calling a Bayes factor a posterior probability, treating credible intervals as confidence intervals, and declaring MCMC convergence from one plot alone. We will deliberately check these points in the questions.

1. Reversing $p(A \mid B)$ and $p(B \mid A)$ in Bayes' theorem.
2. Forgetting the prior when forming the posterior kernel.
3. Forgetting that a posterior must normalize, even when the prior is improper.
4. Mixing Gamma rate and Gamma scale parameterizations.
5. Mixing an Exponential rate with an Exponential mean.
6. Writing a Beta posterior with the wrong number of failures; it is $n - z$, not $z - n$.
7. Using the posterior mode formula when a Beta shape parameter is at most one.
8. Calling the MLE a MAP estimate without checking the prior.
9. Reporting a 95% equal-tail credible interval using the 0.05 and 0.95 quantiles; those quantiles form a 90% interval.

10. Confusing the posterior distribution of a parameter with the posterior predictive distribution of a new observation.
11. Treating a Bayes factor as a posterior probability and ignoring prior model odds.
12. Computing a composite-hypothesis Bayes factor at a point estimate instead of integrating the likelihood over the parameter prior.
13. Omitting the proposal correction in an asymmetric Metropolis–Hastings proposal.
14. Claiming MCMC convergence merely because the trace has many iterations; check mixing, ACF, multiple chains and ESS.
15. Saying ridge, LASSO and spike-and-slab are the same prior; they induce different shrinkage/selection mechanisms.
16. Confusing profile likelihood (maximize nuisance parameter) with Bayesian marginalization (integrate nuisance parameter).

Teacher's spoken explanation

The method is now in place. The remaining work is to practise recognizing which part of the Bayesian story a question belongs to. In the next chapter, the seven question types deliberately move from basic updating to richer models. Read each question as a small scientific investigation: identify the unknown, identify the data model, specify the prior, determine the posterior target, and then calculate only what the question asks. This is how we turn understanding into a reliable examination performance.

Chapter 10

Seven question types: solved examples and practice

Teacher's spoken explanation

Now we will test whether we can tell the story ourselves. Each question type is a new scene in the same investigation. Some questions ask us to update a probability; others change the data type, ask for a summary, compare hypotheses, require a simulation step, or introduce predictors and groups. I will not treat these as seven isolated tricks. For every question, we will first identify what the examiner is asking, connect it to the correct part of the Bayesian workflow, and only then calculate. In the fully solved questions, follow the reasoning as if we were writing on the board. In the practice questions, pause at the stated point and let the student attempt the next step.

Teacher's spoken explanation

We now move from exposition to examination practice. Each question is introduced by identifying the skill it tests and the reasoning a clear written answer should follow. Pause for the student's attempt before revealing the worked solution.

Instructions

Teacher's spoken explanation

Before each question, read the statement slowly and identify the unknown quantity. Ask the student to say aloud which distribution is being used, what the parameter means, and what is being calculated. Then decide whether the task is updating, estimation, model comparison, simulation, regression or hierarchical reasoning.

Each type contains five exam-style questions. Questions 1 and 2 are fully solved. Questions 3, 4 and 5 are left for independent practice here. Their complete, step-by-step solutions are in the next chapter. Unless explicitly stated otherwise, observations are conditionally independent given the model parameters and all Gamma distributions use the shape–rate parameterization.

10.1 Type 1: Bayes' theorem, prior/posterior and prediction

Teacher's spoken explanation

We begin with the first question type because it is the first act of Bayesian reasoning: move from prior uncertainty to posterior uncertainty. Before doing arithmetic, ask what event or parameter is unknown and what has been observed. Then use the same sequence from Chapter 1: prior, likelihood, evidence, posterior, and prediction if requested. These questions are the foundation on which every later type is built.

Teacher's spoken explanation

Let us work through type 1: bayes' theorem, prior/posterior and prediction carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 1 (fully solved): diagnostic testing

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Begin by separating sensitivity from the probability of disease after a positive result. Ask the class to identify the prior prevalence, the true-positive probability and the false-positive probability. On the board, use the law of total probability to calculate the overall positive-test probability, then use Bayes' theorem. Finish by explaining why a rare disease can still have a modest posterior probability despite a good test. The Bayes factor is the ratio of the two conditional positive-test probabilities; it is not the posterior probability. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

A disease affects 2% of a population. A test has sensitivity 0.95 and false-positive probability 0.10. A randomly selected person tests positive. Calculate (i) the probability of a positive result, (ii) the posterior probability that the person has the disease, and (iii) the Bayes factor in favor of disease versus no disease.

Solution. Let D denote disease and T^+ a positive result. The prior probabilities are $P(D) = 0.02$ and $P(D^c) = 0.98$. The likelihoods are $P(T^+ \mid D) = 0.95$ and $P(T^+ \mid D^c) = 0.10$.

(i) By total probability,

$$P(T^+) = 0.02(0.95) + 0.98(0.10) = 0.019 + 0.098 = 0.117.$$

(ii) Bayes' theorem gives

$$P(D \mid T^+) = \frac{0.02(0.95)}{0.117} = \frac{0.019}{0.117} = 0.16239 \text{ (approximately)}.$$

(iii) The Bayes factor for D over D^c is the likelihood ratio

$$\mathrm{BF}_{10} = \frac{P(T^+ \mid D)}{P(T^+ \mid D^c)} = \frac{0.95}{0.10} = 9.5.$$

The factor 9.5 multiplies prior odds; the posterior probability is not 9.5 or 95%. The prior odds are 0.02/0.98, which explains why the posterior probability remains below one fifth.

Question 2 (fully solved): updating prior odds

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Start with the prior odds, not the prior probability. Calculate the likelihood ratio for the observed event, multiply prior odds by that Bayes factor, and convert the resulting odds back to a probability. Pause at the conversion step: odds a to b correspond to probability a divided by a plus b. Emphasize that evidence multiplies odds; it does not multiply a probability directly. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Two hypotheses H_0 and H_1 have prior probabilities $P(H_1) = 0.20$ and $P(H_0) = 0.80$. An observed event E has probabilities $P(E \mid H_1) = 0.60$ and $P(E \mid H_0) = 0.10$. Find the Bayes factor, posterior odds and posterior probability of H_1.

Solution. The Bayes factor is

$$\mathrm{BF}_{10} = \frac{P(E \mid H_1)}{P(E \mid H_0)} = \frac{0.60}{0.10} = 6.$$

Prior odds in favor of H_1 are $0.20/0.80 = 1/4$. Thus posterior odds are

$$\frac{P(H_1 \mid E)}{P(H_0 \mid E)} = 6 \times \frac{1}{4} = \frac{3}{2}.$$

If the odds are $3 : 2$, the posterior probability of H_1 is

$$P(H_1 \mid E) = \frac{3}{3+2} = 0.60.$$

The data multiply the prior odds by six. They do not multiply the prior probability by six.

Question 3 (practice): three-prisoners problem

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. The central issue is the guard's selection rule. Write the three possible pardoned prisoners as separate cases and state the probability that the guard names B under each case. The likelihood of hearing the statement is different in the case where A is pardoned than in the case where B is pardoned. Apply Bayes' theorem using those likelihoods. Conclude by explaining why the rule used to generate the information is part of the probability model. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Three prisoners A, B, C have equal prior probabilities of being pardoned. Prisoner A asks a guard to name one of B, C who will be executed. If both B and C will be executed, the guard chooses between them with equal probability. If the guard says "B will be executed", calculate the conditional probability that A is pardoned. State explicitly the conditional probabilities used and explain why the guard's rule is part of the model.

Question 4 (practice): prior predictive count under a uniform prior

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Condition on theta first. Given theta, the sum of three Bernoulli observations is Binomial with three trials and success probability theta. Then integrate that conditional probability against the Uniform(0,1) prior. Evaluate the beta integral for each possible count z. The result is that each of the four count values has probability one quarter; verify the four probabilities sum to one. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Let $Y_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ for $i = 1, 2, 3$, and let $\theta \sim \text{Uniform}(0, 1)$. Derive the prior predictive distribution of $Z = Y_1 + Y_2 + Y_3$. Calculate $P(Z = 0)$, $P(Z = 1)$, $P(Z = 2)$ and $P(Z = 3)$, and verify that they sum to one.

Question 5 (practice): posterior prediction under a Beta prior

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Count successes and failures, update the two Beta shape parameters, and then use the posterior mean as the probability of success on the next Bernoulli trial. Explain the predictive step by integrating theta against the posterior. The prior Beta(2,3) acts like two prior successes and three prior failures, so the data are added to those pseudo-counts. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

A coin has unknown head probability θ. The prior is $\theta \sim \text{Beta}(2, 3)$. After $n = 7$ tosses, four heads are observed. Find the posterior distribution, posterior mean, and predictive probability that the next toss is a head. Explain the role of the prior pseudo-counts.

10.2 Type 2: discrete conjugate models and exponential-family updates

Teacher's spoken explanation

The first set taught us to update beliefs in general. Now the data have recognizable discrete sampling distributions, so we can exploit conjugacy. The challenge is to notice the sufficient statistic: number of successes, total count and exposure, waiting-time total, or category counts. These summaries carry exactly the information needed for the corresponding posterior update.

Teacher's spoken explanation

Let us work through type 2: discrete conjugate models and exponential-family updates carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the

same one used in the preceding derivation.

Question 1 (fully solved): Beta–Bernoulli update

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Let z be the number of heads and n minus z the number of tails. Write the likelihood as theta to z times one minus theta to n minus z. Multiply by the Beta prior kernel and match powers to identify the posterior Beta parameters. Once the posterior is identified, use its mean, mode and predictive mean formulas. Check that the mode formula is used only when both posterior shape parameters exceed one. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

A coin has unknown head probability θ. The prior is $\theta \sim \text{Beta}(2, 3)$. Eight tosses produce five heads and three tails. Find (i) the posterior distribution, (ii) posterior mean, (iii) posterior mode, and (iv) probability of a head on the next toss.

Solution. There are $z = 5$ successes and $n - z = 3$ failures. With a $\text{Beta}(\alpha, \beta)$ prior, the posterior is $\text{Beta}(\alpha + z, \beta + n - z)$. Therefore

$$\theta \mid y \sim \text{Beta}(2 + 5, 3 + 3) = \text{Beta}(7, 6).$$

The posterior mean is

$$E(\theta \mid y) = \frac{7}{7 + 6} = \frac{7}{13} \approx 0.5385.$$

Since both shape parameters exceed one, the posterior mode is

$$\frac{7 - 1}{7 + 6 - 2} = \frac{6}{11} \approx 0.5455.$$

The next toss is a head with probability equal to the posterior mean:

$$P(\tilde{Y} = 1 \mid y) = E(\theta \mid y) = \frac{7}{13}.$$

The prior behaves like two prior successes and three prior failures. The data add five successes and three failures, giving seven and six in total.

Question 2 (fully solved): Gamma–Poisson update with exposure

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. The exposure values matter. For observations with means t_i times lambda, the likelihood kernel is lambda to the total count multiplied by exp of minus lambda times total exposure. Add the prior Gamma shape to the total count and the prior rate to the total exposure. Then calculate the posterior mean as shape divided by rate. Ask the student to explain why the exposure, not the number of rows alone, appears in the updated rate. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

A Poisson event rate λ has prior $\lambda \sim \text{Gamma}(2, 3)$, where the second parameter is the rate. Counts are observed over exposures $t = (1, 2, 1, 2)$ with values $y = (2, 1, 0, 3)$, and $Y_i \mid \lambda \sim \text{Poisson}(t_i \lambda)$. Find the posterior distribution and posterior mean of λ.

Solution. The total count is $\sum_i y_i = 2+1+0+3 = 6$. The total exposure is $\sum_i t_i = 1+2+1+2 = 6$. The likelihood kernel is

$$L(\lambda \mid y) \propto \lambda^{\sum_i y_i} \exp\left(-\lambda \sum_i t_i\right) = \lambda^6 e^{-6\lambda}.$$

The prior kernel is $\lambda^{2-1}e^{-3\lambda}$. Hence

$$p(\lambda \mid y) \propto \lambda^7 e^{-9\lambda},$$

which is a Gamma density with shape 8 and rate 9:

$$\lambda \mid y \sim \text{Gamma}(8, 9).$$

Its posterior mean is $8/9 \approx 0.8889$. The exposure total, not the number of rows, updates the rate parameter.

Question 3 (practice): Dirichlet–Multinomial

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Treat the category probabilities as a vector that sums to one. Write the Multinomial likelihood kernel as the product of each probability raised to its category count. Multiply by the Dirichlet prior kernel. Matching exponents shows that every prior concentration parameter receives the corresponding observed category count. Use the updated concentration divided by the updated total for the posterior predictive probability of a category. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

A population has three possible categories with probability vector $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$. The prior is Dirichlet$(2, 1, 1)$. A sample of six observations gives category counts $(4, 2, 0)$. Find the posterior distribution, the posterior mean of each category probability, and the predictive probability that the next observation is in category 2.

Question 4 (practice): Gamma–Exponential

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. For Exponential observations parameterized by rate lambda, write the likelihood as lambda to the sample size times exp of minus lambda times the sum of the waiting times. Multiply by the Gamma(shape, rate) prior. The posterior shape is prior shape plus n, and the posterior rate is prior rate plus the observed waiting-time sum. Distinguish this from the Gamma–Poisson model: the data summary that updates the rate is different. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Waiting times $Y_1, \ldots, Y_5$ are iid Exponential with rate λ. The prior is $\lambda \sim \text{Gamma}(2, 1)$ (shape–rate). The observed waiting times sum to 8. Derive the posterior distribution and calculate its mean. State why the likelihood contributes 5 to the shape and 8 to the rate.

Question 5 (practice): Beta–Binomial future count

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. First update the Beta prior using the observed Binomial count. Then introduce the future number of successes conditional on theta, and integrate theta using the Beta posterior. This gives a Beta–Binomial predictive distribution. Explain why the future trials are conditionally independent given theta but become dependent after theta is integrated out. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

A prior $\theta \sim \text{Beta}(1, 1)$ is used for a binomial success probability. Four trials produce three successes. Let K be the number of successes in the next two independent trials. Derive the posterior distribution of θ and calculate $P(K = 2 \mid y)$ using the Beta–Binomial predictive formula.

10.3 Type 3: continuous conjugacy and Normal models

Teacher's spoken explanation

We now move from discrete observations to continuous measurements. The logic has not changed; only the likelihood kernel has changed. Watch how completing the square produces a Normal posterior when the variance is known, and how the Normal–Inverse-Gamma structure handles uncertainty in both the mean vector and variance. The purpose is to recognize the algebra, not merely to memorize the final parameters.

Teacher's spoken explanation

Let us work through type 3: continuous conjugacy and normal models carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 1 (fully solved): Normal–Normal with known variance

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Write the prior and sampling model as Gaussian kernels in the unknown mean. Complete the square in theta. The updated precision is prior precision plus data precision, and the updated mean is the precision-weighted average of the prior mean and sample mean. If there are n observations, use the variance of the sample mean, sigma squared divided by n. Explain the weight of each source of information. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Suppose $Y_i \mid \mu \overset{\text{iid}}{\sim} N(\mu, 4)$ for $i = 1, \ldots, 5$. The prior is $\mu \sim N(6, 9)$. The sample mean is $\overline{y} = 7.2$. Find the posterior distribution of μ, its posterior standard deviation, and the posterior predictive distribution of one new observation.

Solution. The prior precision is $1/9$ and the data precision for the sample mean is $n/\sigma^2 = 5/4$. The posterior precision is

$$\tau_n^{-2} = \frac{1}{9} + \frac{5}{4} = \frac{49}{36}, \qquad \tau_n^2 = \frac{36}{49} \approx 0.7347.$$

The posterior mean is

$$\mu_n = \tau_n^2 \left(\frac{6}{9} + \frac{5(7.2)}{4} \right) = \frac{36}{49} \left(\frac{2}{3} + 9 \right) = \frac{348}{49} \approx 7.1020.$$

Thus $\mu \mid y \sim N(7.1020, 0.7347)$ and posterior standard deviation is $\sqrt{0.7347} \approx 0.8571$. A future observation includes sampling variance 4 in addition to uncertainty about μ:

$$\widetilde{Y} \mid y \sim N(7.1020, 4 + 0.7347) = N(7.1020, 4.7347).$$

Question 2 (fully solved): Normal–Inverse-Gamma update

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. State the conditional Normal prior for the mean given the variance and the inverse-Gamma prior for the variance. Calculate the sample size, sample mean and sum of squares. Update the mean-prior precision parameter, the posterior centre, the inverse-Gamma shape and scale. Take care not to confuse the residual sum of squares around the sample mean with the sum of squared deviations around the updated posterior centre. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Let $Y_i \mid \mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ for data $4, 5, 6, 7$. Use the prior $\mu \mid \sigma^2 \sim N(5, \sigma^2/2)$ and $\sigma^2 \sim \text{InvGamma}(3, 2)$ (shape–scale). Find the posterior hyperparameters and the posterior predictive distribution of a future observation.

Solution. Here $n = 4$, $\overline{y} = 5.5$, $S = \sum (y_i - \overline{y})^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5$, $\mu_0 = 5$, $\kappa_0 = 2$, $a_0 = 3$, and $b_0 = 2$. Update

$$\kappa_n = 2 + 4 = 6, \qquad \mu_n = \frac{2(5) + 4(5.5)}{6} = \frac{32}{6} = \frac{16}{3}.$$

Also

$$a_n = 3 + \frac{4}{2} = 5,$$

$$b_n = 2 + \frac{5}{2} + \frac{2(4)}{2(6)}(5.5 - 5)^2 = 2 + 2.5 + \frac{8}{12}(0.25) = \frac{14}{3} \approx 4.6667.$$

Therefore

$$\sigma^2 \mid y \sim \text{InvGamma}\left(5, \frac{14}{3}\right), \quad \mu \mid \sigma^2, y \sim N\left(\frac{16}{3}, \frac{\sigma^2}{6}\right).$$

The predictive distribution is Student-t with $2a_n = 10$ degrees of freedom, location $\mu_n = 16/3$, and scale

$$s_* = \sqrt{\frac{b_n(\kappa_n + 1)}{a_n \kappa_n}} = \sqrt{\frac{(14/3)7}{5(6)}} = \sqrt{\frac{49}{45}} \approx 1.0435.$$

Hence $\widetilde{Y} \mid y \sim t_{10}(16/3, 1.0435)$ under the scale convention stated in Chapter 2.

Question 3 (practice): known mean, unknown variance

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Because the mean is known, the likelihood for the variance depends on the sum of squared deviations from that known mean. Combine this kernel with the inverse-Gamma prior and match the power of the variance and its reciprocal-exponential term. The posterior remains inverse-Gamma. Highlight that using the sample mean in the sum of squares would be incorrect because the mean is specified as known. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Suppose $Y_1, \ldots, Y_5 \mid \sigma^2 \overset{\text{iid}}{\sim} N(2, \sigma^2)$, where the mean 2 is known. The prior is $\sigma^2 \sim \text{Inv-}\chi^2(4, 9)$. The observed sum $\sum_i (y_i - 2)^2$ is 10. Find the posterior degrees of freedom and scale, and state the posterior predictive distribution of one new observation.

Question 4 (practice): unknown mean and variance with an improper prior

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Start with the joint likelihood for the Normal sample and the stated improper prior. Integrate out the unknown mean or complete the square in it. The remaining kernel in the variance identifies the posterior distribution. Then check the degrees of freedom and propriety conditions. Explain why an improper prior is acceptable only when the resulting posterior can be normalized. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Let $Y_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ and use the prior kernel $\pi(\mu, \sigma^2) \propto 1/\sigma^2$. The data are $1, 2, 3, 4, 5$. Derive $\mu \mid \sigma^2, y$ and $\sigma^2 \mid y$, calculate the parameters of both distributions, and verify that the posterior is proper.

Question 5 (practice): NIG posterior predictive distribution

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Do not stop after finding the posterior for the parameters. For a future observation, integrate the Normal sampling distribution over both the coefficient or mean uncertainty and variance uncertainty. Under the Normal–Inverse-Gamma model, the result is a Student t predictive distribution. Identify its location, scale and degrees of freedom and explain why it has heavier tails than a Normal plug-in prediction. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Suppose $Y_i \mid \mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ and the prior is $\mu \mid \sigma^2 \sim N(0, \sigma^2/1)$ and $\sigma^2 \sim \text{InvGamma}(2, 1)$. The data are $-1, 0, 1$. Calculate $(\kappa_n, \mu_n, a_n, b_n)$ and give the full posterior predictive Student-t distribution for a new observation. Do not stop at the conditional Normal posterior for μ.

10.4 Type 4: posterior summaries, MAP and credible intervals

Teacher's spoken explanation

A posterior distribution is the complete result, but an examination may ask us to report a point or a range. These questions ask us to extract that information carefully. Always ask which summary is requested and why: a posterior mean, a median, a mode, an equal-tail interval, an HPD interval, or an action chosen under a stated loss function.

Teacher's spoken explanation

Let us work through type 4: posterior summaries, map and credible intervals carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 1 (fully solved): Beta posterior summaries

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. After updating the Beta parameters, calculate the posterior mean, variance or standard deviation, and mode where it exists. Compare the posterior mean and mode, and then compute the predictive success probability using the posterior mean. State the shape restrictions before using the interior mode formula; boundary cases require separate treatment. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

A coin has a uniform prior $\theta \sim \text{Beta}(1,1)$. Five tosses produce two heads. Find the posterior distribution, posterior mean, posterior mode, MLE, posterior variance, and explain which estimate is selected under squared-error loss.

Solution. Here $z = 2$ and $n - z = 3$. The posterior is $\text{Beta}(3,4)$. Its posterior mean is $3/(3+4) = 3/7$. Since both shapes exceed one, the mode is

$$\frac{3-1}{3+4-2} = \frac{2}{5} = 0.4.$$

The MLE is $z/n = 2/5 = 0.4$, so MAP and MLE coincide under the uniform prior in this example. The posterior variance is

$$\text{Var}(\theta \mid y) = \frac{3(4)}{7^2(8)} = \frac{12}{392} = \frac{3}{98} \approx 0.03061.$$

Squared-error loss selects the posterior mean $3/7$, not the posterior mode. The posterior mean uses all posterior mass, while the MAP selects the density maximum.

Question 2 (fully solved): Normal posterior and a 95% credible interval

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. We will treat (fully solved): normal posterior and a 95

Let $Y_i \mid \mu \overset{\text{iid}}{\sim} N(\mu, 1)$ with $n = 4$ and $\overline{y} = 2$. The prior is $\mu \sim N(0, 4)$. Find the posterior distribution and its central 95% credible interval. Also give the predictive distribution for a new observation.

Solution. Prior precision is $1/4$ and data precision is $n/\sigma^2 = 4$. Thus

$$\tau_n^2 = (1/4 + 4)^{-1} = \frac{4}{17} \approx 0.23529,$$

$$\mu_n = \tau_n^2(0/4 + 4 \cdot 2) = \frac{4}{17}(8) = \frac{32}{17} \approx 1.88235.$$

Therefore $\mu \mid y \sim N(32/17, 4/17)$ and posterior standard deviation is $\sqrt{4/17} \approx 0.48507$. The central 95% equal-tail interval is

$$\mu_n \pm 1.96\sqrt{\tau_n^2} = 1.88235 \pm 1.96(0.48507) \approx [0.9316, 2.8331].$$

For a new observation, add the sampling variance 1 to the posterior variance of the mean:

$$\widetilde{Y} \mid y \sim N\left(\frac{32}{17}, 1 + \frac{4}{17}\right) = N\left(1.88235, \frac{21}{17}\right).$$

Question 3 (practice): MAP under a transformation

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. This problem is testing the Jacobian. Transform the posterior density into the new parameter, including the absolute derivative of the inverse transformation. Then maximize the transformed density. Do not simply transform the old MAP estimate: density maxima are not invariant under nonlinear reparameterization. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Suppose a posterior density for $\theta > 0$ is Gamma$(2, 1)$ with shape 2 and rate 1, so $p_\theta(\theta \mid y) = \theta e^{-\theta}$ for $\theta > 0$. Find the MAP estimate of θ. Now set $\psi = \log \theta$, derive the transformed posterior density and find its MAP estimate. Explain why the transformed MAP is not the logarithm of the original MAP.

Question 4 (practice): equal-tail Beta credible interval

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Identify the posterior Beta parameters, then define the lower and upper posterior quantiles. For a 95 percent equal-tail interval, use the 0.025 and 0.975 quantiles. Explain that these quantiles are generally obtained numerically or from a Beta quantile function; there is no requirement that the interval be symmetric around the posterior mean. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

After observing data, a parameter has posterior distribution $\theta \mid y \sim \text{Beta}(3, 4)$. Construct a central 90% equal-tail credible interval. Use the Beta quantiles $q_{0.05} = 0.1532$ and $q_{0.95} = 0.7287$ (rounded to four decimals). State the posterior probability represented by the interval and distinguish it from a confidence interval statement.

Question 5 (practice): point estimate under asymmetric loss

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Write the loss function before selecting an estimator. Under asymmetric absolute loss, errors on one side cost more than errors on the other, so the posterior median is replaced by a posterior quantile determined by the relative loss weights. Differentiate the posterior expected loss or use the quantile characterization, then interpret why the estimate shifts toward the less costly side. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Suppose the posterior distribution is $\theta \mid y \sim \text{Beta}(4, 2)$. A decision a is penalized by $3(\theta - a)$ when $a < \theta$ (underestimation) and by $(a - \theta)$ when $a > \theta$ (overestimation). Show that the Bayes action is the 0.75 posterior quantile and write it in terms of the Beta quantile function. Explain why it is not generally the posterior mean or median.

10.5 Type 5: Bayesian hypothesis testing, Bayes factors and prior choice

Teacher's spoken explanation

We now change from describing a parameter to comparing claims. In simple hypotheses, the Bayes factor is a likelihood ratio. In composite hypotheses, we must integrate over the parameter under each model. The prior enters that integral, so these questions connect directly back to the prior-choice chapter. Keep prior odds, Bayes factor and posterior odds as three separate quantities.

Teacher's spoken explanation

Let us work through type 5: bayesian hypothesis testing, bayes factors and prior choice carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 1 (fully solved): simple-hypothesis Bayes factor

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Each simple hypothesis fixes the parameter at one value, so its marginal likelihood is simply the likelihood at that value. Divide the likelihood under the alternative by the likelihood under the null to obtain the Bayes factor. Multiply by prior odds to obtain posterior odds, and only then normalize to get a posterior probability. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Let $Y_1, \ldots, Y_5$ be iid Bernoulli(p). Test $H_0 : p = 0.10$ against $H_1 : p = 0.50$. The data contain four successes and one failure. The prior model probabilities are $P(H_1) = 0.25$ and $P(H_0) = 0.75$. Find the Bayes factor and posterior probability of H_1.

Solution. The binomial coefficient is common to both hypotheses and cancels. The likelihoods are

$$p(y \mid H_1) \propto 0.5^4(0.5)^1 = 0.5^5 = 0.03125,$$

$$p(y \mid H_0) \propto 0.1^4(0.9) = 0.00009.$$

Thus

$$\mathrm{BF}_{10} = \frac{0.03125}{0.00009} \approx 347.2222.$$

Prior odds $H_1 : H_0$ are $0.25/0.75 = 1/3$, so posterior odds are approximately $115.7407 : 1$. Hence

$$P(H_1 \mid y) = \frac{115.7407}{1 + 115.7407} \approx 0.9914.$$

The evidence strongly favors H_1, and the prior model odds have been multiplied by the Bayes factor.

Question 2 (fully solved): Bayes factor for a composite Normal alternative

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Under the composite alternative, integrate the likelihood over the prior distribution on the unknown parameter. This is the marginal likelihood; it is not the likelihood evaluated at the prior mean or at the MLE. Compare it with the null likelihood to form the Bayes factor. Explain that averaging over a range of parameter values creates an automatic complexity penalty when much of the prior mass predicts the data poorly. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Suppose $Y_i \mid \mu \overset{\text{iid}}{\sim} N(\mu, 1)$, $n = 4$, and $\overline{y} = 0.75$. Compare $H_0 : \mu = 0$ with $H_1 : \mu \sim N(0, 4)$. Use equal prior model probabilities. Calculate BF_{10} using the marginal distribution of $\overline{Y}$ and then calculate $P(H_1 \mid y)$.

Solution. Under H_0, $\overline{Y} \sim N(0, 1/4)$. Under H_1, integrating over $\mu \sim N(0, 4)$ gives

$$\overline{Y} \mid H_1 \sim N\left(0, 4 + \frac{1}{4}\right) = N(0, 4.25).$$

The residual component of the Normal sample likelihood is the same under both models and cancels in the ratio. Therefore

$$\mathrm{BF}_{10} = \frac{\phi(0.75; 0, 4.25)}{\phi(0.75; 0, 0.25)} = \sqrt{\frac{0.25}{4.25}} \exp\left\{-\frac{0.75^2}{2(4.25)} + \frac{0.75^2}{2(0.25)}\right\} \approx 0.6992.$$

Since prior model probabilities are equal, prior odds are one and posterior odds equal $0.6992 : 1$. Thus

$$P(H_1 \mid y) = \frac{0.6992}{1 + 0.6992} \approx 0.4115.$$

The observed sample mean is close enough to zero that the point null has slightly greater marginal predictive density. Notice that the alternative is composite: its likelihood is averaged over the prior $N(0, 4)$.

Question 3 (practice): Jeffreys' prior for a Poisson rate

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Compute the Fisher information for the Poisson rate from the score or the expected negative second derivative. Take its square root to obtain the Jeffreys prior kernel. State whether the resulting prior is proper on the positive real line. The posterior can still be proper after data, but do not confuse those two questions. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

For $Y \sim \text{Poisson}(\lambda)$, derive the Fisher information for one observation and hence derive Jeffreys' prior. If n independent observations have total count s, state the posterior under this prior, using the Gamma shape–rate parameterization. Is the prior itself proper?

Question 4 (practice): Jeffreys' prior for a Bernoulli probability

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. For Bernoulli data, derive Fisher information for theta and take its square root. The resulting density is proportional to theta to minus one half times one minus theta to minus one half, which is the Beta one-half, one-half prior. It favours values near the boundaries more than a uniform prior does, while remaining proper. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Let $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, and use Jeffreys' prior. If $n = 10$ observations contain $z = 7$ successes, derive the posterior distribution and calculate the posterior predictive probability of success on the next observation.

Question 5 (practice): prior odds are not the Bayes factor

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Use the numbers to calculate prior odds and the Bayes factor separately. Then multiply them to obtain posterior odds. The key conceptual correction is that a Bayes factor measures the data's relative support for two models; prior odds express the starting preference. A Bayes factor cannot be read as the probability that a model is true. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

A Bayes factor in favor of H_1 over H_0 is 4. The prior probability of H_1 is 0.10. Calculate prior odds, posterior odds and posterior probability of H_1. Explain why it is incorrect to report 0.4 as the posterior probability.

10.6 Type 6: MCMC, Gibbs sampling, Metropolis–Hastings and diagnostics

Teacher's spoken explanation

These questions begin where algebraic conjugacy may end. We are given a posterior target and must reason about a chain designed to explore it. A numerical acceptance probability is only one part of the answer; the student must also understand proposal asymmetry, conditional updates, mixing and what a diagnostic plot can or cannot establish.

Teacher's spoken explanation

Let us work through type 6: mcmc, gibbs sampling, metropolis–hastings and diagnostics carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 1 (fully solved): Metropolis–Hastings acceptance

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Record the current state, proposed state, target density values and proposal probabilities in a small table. Substitute them into the full Metropolis–Hastings ratio. Take the minimum of one and that ratio. If the proposal is accepted, move; if rejected, remain at the current state. Explain that the stay-put repetition is a legitimate part of the Markov chain. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The target density is $\pi(\theta) \propto e^{-\theta^2/2}$ on the real line. The current state is $\theta = 0.5$ and a symmetric proposal suggests $\theta^* = 1.2$. Calculate the MH acceptance probability. If the random number used for the accept/reject step is $u = 0.31$, does the chain accept the proposal?

Solution. The proposal is symmetric, so the proposal ratio is one. The acceptance probability is

$$\alpha = \min\left\{1, \frac{e^{-1.2^2/2}}{e^{-0.5^2/2}}\right\} = \min\{1, e^{-(1.44-0.25)/2}\} = e^{-0.595} \approx 0.5516.$$

The chain accepts when $u \le \alpha$. Since $0.31 < 0.5516$, the proposal is accepted and the next state is 1.2. Had u exceeded 0.5516, the chain would remain at 0.5.

Question 2 (fully solved): one Gibbs sweep

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. List the current values of every component. Then update the first component from its full conditional distribution, using the current values of all other components. Move to the next component and condition on the most recently updated values. One complete pass through all components is one Gibbs sweep. Emphasize that Gibbs uses exact full conditional draws, so there is no separate acceptance-rejection step. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Let (X, Y) be bivariate Normal with mean zero, unit marginal variances and correlation $\rho = 0.8$. Its full conditionals are $Y \mid X = x \sim N(0.8x, 0.36)$ and $X \mid Y = y \sim N(0.8y, 0.36)$. Starting from $X^{(0)} = 1$, suppose the two standard Normal draws used in one Gibbs sweep are $z_1 = 0.5$ for updating Y and $z_2 = -1$ for updating X. Calculate $Y^{(1)}$ and $X^{(1)}$.

Solution. The conditional standard deviation is $\sqrt{0.36} = 0.6$. First update Y using the latest $X^{(0)} = 1$:

$$Y^{(1)} = 0.8(1) + 0.6(0.5) = 0.8 + 0.3 = 1.1.$$

Then update X using the new $Y^{(1)} = 1.1$:

$$X^{(1)} = 0.8(1.1) + 0.6(-1) = 0.88 - 0.6 = 0.28.$$

The second update must use the newly sampled $Y^{(1)}$, not the old $Y^{(0)}$. The samples are dependent, but the target bivariate Normal is the stationary distribution of this Gibbs transition under the usual regularity conditions.

Question 3 (practice): diagnose a trace plot

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Describe what the plotted sequence does over iteration. Look for a stable horizontal band, frequent movement and no long-term trend. A drift, abrupt regime change or long flat segments suggests poor mixing or failure to reach stationarity. A single trace plot is not enough: mention multiple chains, density overlays and autocorrelation as complementary checks. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Three MCMC chains are run for a scalar parameter. Chain 1 begins near -4 and stays near -4; Chain 2 begins near 0 and stays near 0; Chain 3 begins near 4 and stays near 4. Each chain has 20,000 iterations and appears visually flat. Explain why this is not evidence of convergence. Name at least four additional graphical or numerical diagnostics that should be examined and state what a healthy pattern looks like.

Question 4 (practice): asymmetric Metropolis–Hastings proposal

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Use the complete acceptance ratio and retain the proposal-density correction. Write q(current given proposed) in the numerator and q(proposed given current) in the denominator.

Only cancel these terms when the proposal is symmetric. This question checks whether the student understands the Hastings correction rather than memorizing the random-walk special case. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

A target distribution on states $\{0, 1\}$ has $\pi(0) = 0.6$ and $\pi(1) = 0.4$. A proposal has $q(1 \mid 0) = 0.8$ and $q(0 \mid 1) = 0.2$. From state 0, the chain proposes state 1. Calculate the acceptance probability. Explain what the answer would be if the proposal ratio were incorrectly omitted.

Question 5 (practice): effective sample size

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Start from the stored number of MCMC draws and the autocorrelation at successive lags. Use the integrated autocorrelation time to estimate effective sample size. Then calculate the Monte Carlo standard error of the posterior mean as the posterior standard deviation divided by the square root of ESS. Explain why ten thousand correlated draws can contain far less information than ten thousand independent draws. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

A stationary MCMC output contains $M = 1000$ draws. Approximate its autocorrelation by an AR(1) pattern $\rho_k = 0.8^k$. Use the approximation $\text{ESS} \approx M(1 - \rho_1)/(1 + \rho_1)$ to estimate the effective sample size. If the posterior standard deviation is 2, estimate the Monte Carlo standard error of the sample mean and compare it with the standard error from 1000 independent draws.

10.7 Type 7: Bayesian regression, regularization and hierarchical models

Teacher's spoken explanation

The final question type brings together several themes. Regression extends the parameter from a scalar to a vector. Priors control coefficient size and can express uncertainty about variable inclusion. Hierarchical models then recognize that observations may be grouped and that group-specific parameters should share information. Treat this as the synthesis set: it asks whether you can move from a simple model to a structured model without losing the Bayesian logic.

Teacher's spoken explanation

Let us work through type 7: bayesian regression, regularization and hierarchical models carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 1 (fully solved): ridge regression as a Normal-prior MAP

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Write the Gaussian likelihood and a zero-centred multivariate Normal prior on the coefficients. Take the negative log posterior and discard constants that do not depend on beta. The remaining expression is residual sum of squares plus a multiple of the squared coefficient norm. Identify the ridge penalty and explain how prior variance controls shrinkage. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Consider the no-intercept model $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, I_2)$, $X = (1, 2)^\mathsf{T}$, and $y = (1, 2)^\mathsf{T}$. Put the prior $\beta \sim N(0, 1)$. Find the posterior distribution and show that the MAP estimate is the ridge solution.

Solution. Here $X^\mathsf{T}X = 1^2 + 2^2 = 5$ and $X^\mathsf{T}y = 1(1) + 2(2) = 5$. Since $\sigma^2 = 1$, $V_0 = 1$ and $\beta_0 = 0$,

$$V_n = (1 + 5)^{-1} = \frac{1}{6}, \qquad \beta_n = \frac{1}{6}(5) = \frac{5}{6}.$$

Thus $\beta \mid y \sim N(5/6, 1/6)$ and its mode is $5/6$. The negative log posterior, up to constants, is

$$\frac{1}{2}\|y - X\beta\|^2 + \frac{1}{2}\beta^2.$$

Multiplying by two gives the ridge objective $\|y - X\beta\|^2 + \beta^2$. Differentiating gives $(X^\mathsf{T}X + 1)\widehat{\beta} = X^\mathsf{T}y$, so $\widehat{\beta} = 5/6$. The Gaussian prior is the Bayesian source of the L_2 penalty.

Question 2 (fully solved): partial pooling in a Normal hierarchical model

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. For each group, combine its sample information with the Normal prior centred at the population mean. The posterior group mean is a weighted average of the group estimate and the population centre. Compute the group-specific weight from the group sample size and variances. Explain why a small group is shrunk more than a large, precise group. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

For groups $j = 1, 2$, suppose $\overline{Y}_j \mid \theta_j \sim N(\theta_j, \sigma^2/n_j)$ and $\theta_j \sim N(\mu, \tau^2)$, with known $\mu = 10$, $\tau^2 = 4$ and $\sigma^2 = 16$. Group 1 has $n_1 = 2, \overline{y}_1 = 8$ and Group 2 has $n_2 = 8, \overline{y}_2 = 12$. Find each posterior mean and explain the amount of shrinkage.

Solution. For each group,

$$\mathbb{E}(\theta_j \mid \overline{y}_j) = \frac{n_j\tau^2}{n_j\tau^2 + \sigma^2}\overline{y}_j + \frac{\sigma^2}{n_j\tau^2 + \sigma^2}\mu.$$

For Group 1, the data weight is $2(4)/(2(4) + 16) = 8/24 = 1/3$. Hence

$$E(\theta_1 \mid \overline{y}_1) = \frac{1}{3}(8) + \frac{2}{3}(10) = \frac{28}{3} \approx 9.333.$$

For Group 2, the data weight is $8(4)/(8(4) + 16) = 32/48 = 2/3$. Hence

$$E(\theta_2 \mid \overline{y}_2) = \frac{2}{3}(12) + \frac{1}{3}(10) = \frac{34}{3} \approx 11.333.$$

Group 1 has fewer observations and is shrunk more strongly toward the population mean 10. Group 2 has more information and is closer to its own sample mean. This is partial pooling.

Question 3 (practice): derive the LASSO prior

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Begin with the Laplace density and take its negative logarithm. The result is a constant plus an absolute-value penalty. Combine this with the Gaussian regression negative log likelihood to recover the LASSO objective. Explain why a Laplace prior creates a cusp at zero and encourages sparse MAP estimates. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

For the Gaussian regression model $Y \mid \beta, \sigma^2 \sim N(X\beta, \sigma^2 I)$ with known σ^2, suppose the independent prior on each coefficient is Laplace with density

$$\pi(\beta_j \mid \lambda, \sigma^2) = \frac{\lambda}{2\sigma^2} \exp\left(-\frac{\lambda|\beta_j|}{\sigma^2}\right).$$

Derive the negative log posterior up to a constant and show the corresponding MAP optimization problem. Explain why the L_1 penalty can produce exact zeros.

Question 4 (practice): spike-and-slab posterior inclusion

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. Write the prior inclusion probability and the two coefficient distributions: a point mass at zero for the spike and a continuous distribution for the slab. Calculate the marginal likelihood under inclusion and exclusion, then apply Bayes' theorem to the indicator. The final posterior inclusion probability is a probability about whether the variable is in the model, not merely whether its fitted coefficient is numerically large. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

For one coefficient, let $\gamma \sim \text{Bernoulli}(\pi)$ with prior inclusion probability $\pi = 0.20$. The marginal likelihood of the observed data under inclusion is $m_1(y) = 0.12$ and under exclusion is $m_0(y) = 0.03$. Calculate the Bayes factor and posterior inclusion probability $P(\gamma = 1 \mid y)$. State how the spike-and-slab prior encodes exclusion.

Question 5 (practice): empirical Bayes and profile likelihood

Teacher's spoken explanation

Teacher: Read the question aloud and ask the student to identify the model before looking at the answer. First integrate out the group-level effects to obtain the marginal likelihood for the hyperparameters. Empirical Bayes estimates those hyperparameters by maximizing this marginal likelihood. For profile likelihood, instead fix the parameter of interest and maximize over nuisance parameters. Emphasize the distinction: marginal likelihood integrates; profile likelihood maximizes. They are not interchangeable operations. After the independent attempt, turn to the matching worked solution at the end of the manual. Reconstruct every line with the student and ask them to explain the final result in a complete sentence.

Suppose independent group means satisfy

$$\overline{Y}_j \mid \mu, \tau^2 \sim N\left(\mu, \tau^2 + \frac{\sigma^2}{n_j}\right),$$

where σ^2 is known. (i) Write the marginal likelihood for (μ, τ^2) based on observed group means $\overline{y}_1, \ldots, \overline{y}_J$. (ii) For fixed τ^2, derive the maximum-likelihood estimate $\widehat{\mu}(\tau^2)$. (iii) Write the profile likelihood for τ^2. (iv) Explain how empirical Bayes uses the resulting estimate and how this differs from fully Bayesian hierarchical inference.

Teacher's spoken explanation

We have now worked through the seven families of examination questions. Notice how the reasoning has travelled across the full course: posterior updating, conjugacy, continuous models, estimation, testing and priors, simulation, regression and hierarchical structure. The practice problems were not separate from the theory; they were the theory in action. We will finish by returning to every practice question and showing a complete solution. As you listen, compare the solution with your own attempt and identify not only whether the answer is correct, but whether the reasoning is complete.

Chapter 11

Complete solutions to the 21 practice questions

Teacher's spoken explanation

We now return to the 21 questions that were left for independent practice. Think of this chapter as the second pass through the same lecture. The first pass asked you to make a mathematical decision without seeing the destination. This pass shows how a careful solution is constructed and why each step is necessary. A good solution does not merely state the final distribution: it identifies the model, writes the likelihood, combines it with the prior, handles constants correctly, names the posterior parameters and interprets the requested quantity. As we move through the solutions, notice how the same structure repeats even though the distributions change.

Teacher's spoken explanation

This chapter contains full worked solutions to the 21 questions previously labelled for practice. Do not begin by reading the final answer: reconstruct the model, select the formula, substitute the values, and interpret the result.

This chapter contains the worked solutions to Questions 3–5 in each of the seven types. All notation and assumptions are restated where needed so that the solution can be read independently.

11.1 Solutions: Type 1

Teacher's spoken explanation

Let us work through solutions: type 1 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 3: three-prisoners problem

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. The central issue is the guard's selection rule. Write the three possible pardoned prisoners as separate cases and state the probability that the guard names B under each case. The likelihood of hearing the statement is different in the case where A is pardoned than in the case where B is pardoned. Apply Bayes' theorem using those likelihoods. Conclude by explaining why the rule used to generate the information is part of the probability model. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Let A, B, C denote the events that the named prisoner is pardoned, and let G_B denote the event that the guard says B will be executed. The prior probabilities are each $1/3$. Under the rule in the question:

$$P(G_B \mid A) = \frac{1}{2}, \qquad P(G_B \mid B) = 0, \qquad P(G_B \mid C) = 1.$$

The probability that the guard says B is

$$P(G_B) = \frac{1}{3}\frac{1}{2} + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}.$$

By Bayes' theorem,

$$P(A \mid G_B) = \frac{P(G_B \mid A)P(A)}{P(G_B)} = \frac{(1/2)(1/3)}{1/2} = \frac{1}{3}.$$

The answer remains $1/3$. The guard's reporting rule determines $P(G_B \mid A)$ and $P(G_B \mid C)$. A different guard policy would define a different likelihood and could change the answer.

Question 4: prior predictive count under a uniform prior

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Condition on theta first. Given theta, the sum of three Bernoulli observations is Binomial with three trials and success probability theta. Then integrate that conditional probability against the Uniform(0,1) prior. Evaluate the beta integral for each possible count z. The result is that each of the four count values has probability one quarter; verify the four probabilities sum to one. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

For $Z = Y_1 + Y_2 + Y_3$, conditional on θ, $Z \mid \theta \sim \text{Binomial}(3, \theta)$. The prior is uniform, so

$$P(Z = z) = \binom{3}{z} \int_0^1 \theta^z (1-\theta)^{3-z} \, \mathrm{d}\theta = \binom{3}{z} B(z+1, 4-z).$$

Using $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$,

$$P(Z = z) = \binom{3}{z} \frac{z!(3-z)!}{4!} = \frac{3!}{z!(3-z)!} \frac{z!(3-z)!}{4!} = \frac{1}{4}.$$

Therefore $P(Z = 0) = P(Z = 1) = P(Z = 2) = P(Z = 3) = 1/4$, and the probabilities sum to $4(1/4) = 1$. The uniformity is for the count Z, not for the eight possible ordered sequences.

Question 5: posterior prediction under a Beta prior

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Count successes and failures, update the two Beta shape parameters, and then use the posterior mean as the probability of success on the next Bernoulli trial. Explain the predictive step by integrating theta against the posterior. The prior Beta(2,3) acts like two prior successes and three prior failures, so the data are added to those pseudo-counts. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The prior is Beta$(2, 3)$; four heads and three tails give $z = 4, n - z = 3$. Conjugacy gives

$$\theta \mid y \sim \text{Beta}(2 + 4, 3 + 3) = \text{Beta}(6, 6).$$

The posterior mean is $6/(6 + 6) = 1/2$. Since the next toss is conditionally Bernoulli(θ),

$$P(\widetilde{Y} = 1 \mid y) = E(\theta \mid y) = \frac{1}{2}.$$

The prior contributes two pseudo-successes and three pseudo-failures. The sample contributes four successes and three failures, producing posterior pseudo-count parameters six and six. The posterior predictive probability equals the posterior mean, not the raw sample proportion $4/7$.

11.2 Solutions: Type 2

Teacher's spoken explanation

Let us work through solutions: type 2 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 3: Dirichlet–Multinomial

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Treat the category probabilities as a vector that sums to one. Write the Multinomial likelihood kernel as the product of each probability raised to its category count. Multiply by the Dirichlet prior kernel. Matching exponents shows that every prior concentration parameter receives the corresponding observed category count. Use the updated concentration divided by the updated total for the posterior predictive probability of a category. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The prior is Dirichlet$(2, 1, 1)$ and the observed counts are $(4, 2, 0)$. Add counts componentwise:

$$\boldsymbol{\theta} \mid y \sim \text{Dirichlet}(2 + 4, 1 + 2, 1 + 0) = \text{Dirichlet}(6, 3, 1).$$

The posterior total concentration is $6 + 3 + 1 = 10$. The posterior means are

$$E(\theta_1 \mid y) = \frac{6}{10} = 0.6, \quad E(\theta_2 \mid y) = \frac{3}{10} = 0.3, \quad E(\theta_3 \mid y) = \frac{1}{10} = 0.1.$$

The posterior predictive probability of category 2 on the next draw is 0.3. This is also the posterior mean of θ_2.

Question 4: Gamma–Exponential

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. For Exponential observations parameterized by rate lambda, write the likelihood as lambda to the sample size times exp of minus lambda times the sum of the waiting times. Multiply by the Gamma(shape, rate) prior. The posterior shape is prior shape plus n, and the posterior rate is prior rate plus the observed waiting-time sum. Distinguish this from the Gamma–Poisson model: the data summary that updates the rate is different. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

For an Exponential rate parameter, the likelihood kernel for five observations is

$$L(\lambda \mid y) \propto \lambda^5 \exp\left(-\lambda \sum_i y_i\right) = \lambda^5 e^{-8\lambda}.$$

The Gamma$(2,1)$ prior kernel is $\lambda^1 e^{-\lambda}$. Thus

$$p(\lambda \mid y) \propto \lambda^6 e^{-9\lambda}, \quad \lambda \mid y \sim \text{Gamma}(7,9).$$

Its mean is $7/9 \approx 0.7778$. The five independent Exponential likelihood factors each contribute one power of λ, hence the shape increment is five. Their exponentials combine to $e^{-\lambda \sum_i y_i}$, hence the rate increment is the total waiting time eight.

Question 5: Beta–Binomial future count

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. First update the Beta prior using the observed Binomial count. Then introduce the future number of successes conditional on theta, and integrate theta using the Beta posterior. This gives a Beta–Binomial predictive distribution. Explain why the future trials are conditionally independent given theta but become dependent after theta is integrated out. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

After three successes and one failure, the Beta$(1,1)$ prior gives

$$\theta \mid y \sim \text{Beta}(4,2).$$

For two future trials, $K \mid \theta \sim \text{Binomial}(2,\theta)$. Hence

$$P(K=2 \mid y) = \binom{2}{2} \frac{B(4+2, 2+0)}{B(4,2)} = \frac{B(6,2)}{B(4,2)}.$$

Using $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$,

$$B(6,2) = \frac{5!1!}{7!} = \frac{1}{42}, \qquad B(4,2) = \frac{3!1!}{5!} = \frac{1}{20}.$$

Therefore $P(K=2 \mid y) = (1/42)/(1/20) = 20/42 = 10/21 \approx 0.4762$. Equivalently, this is $E(\theta^2 \mid y)$ because both future trials must be successes.

11.3 Solutions: Type 3

Teacher's spoken explanation

Let us work through solutions: type 3 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 3: known mean, unknown variance

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Because the mean is known, the likelihood for the variance depends on the sum of squared deviations from that known mean. Combine this kernel with the inverse-Gamma prior and match the power of the variance and its reciprocal-exponential term. The posterior remains inverse-Gamma. Highlight that using the sample mean in the sum of squares would be incorrect because the mean is specified as known. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

For $n = 5$ observations from $N(2, \sigma^2)$ and prior $\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, s_0^2)$ with $\nu_0 = 4, s_0^2 = 9$, the posterior degrees of freedom are

$$\nu_n = \nu_0 + n = 4 + 5 = 9.$$

The posterior scale is

$$s_n^2 = \frac{\nu_0 s_0^2 + \sum_i (y_i - 2)^2}{\nu_0 + n} = \frac{4(9) + 10}{9} = \frac{46}{9} \approx 5.1111.$$

Thus $\sigma^2 \mid y \sim \text{Inv-}\chi^2(9, 46/9)$. Integrating the future Normal observation over this variance posterior gives

$$\frac{\widetilde{Y} - 2}{\sqrt{46/9}} \mid y \sim t_9.$$

The predictive distribution is t_9 with location 2 and scale $\sqrt{46/9} \approx 2.2608$.

Question 4: unknown mean and variance with an improper prior

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Start with the joint likelihood for the Normal sample and the stated improper prior. Integrate out the unknown mean or complete the square in it. The remaining kernel in the variance identifies the posterior distribution. Then check the degrees of freedom and propriety conditions. Explain why an improper prior is acceptable only when the resulting posterior can be normalized. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The data are $1, 2, 3, 4, 5$, so $n = 5$, $\overline{y} = 3$, and

$$\text{SSE} = \sum_i (y_i - 3)^2 = 4 + 1 + 0 + 1 + 4 = 10.$$

Under $\pi(\mu, \sigma^2) \propto 1/\sigma^2$, the conditional posterior is

$$\mu \mid \sigma^2, y \sim N\left(\overline{y}, \frac{\sigma^2}{n}\right) = N\left(3, \frac{\sigma^2}{5}\right).$$

The variance posterior is

$$\sigma^2 \mid y \sim \text{InvGamma}\left(\frac{n-1}{2}, \frac{\text{SSE}}{2}\right) = \text{InvGamma}(2, 5).$$

Equivalently, $\sigma^2 \mid y \sim \text{Inv-}\chi^2(4, s^2)$ with $s^2 = \text{SSE}/(n-1) = 10/4 = 2.5$. The posterior is proper because $n > 1$ and $\text{SSE} > 0$; the Normal conditional is proper and the inverse-Gamma variance posterior has positive shape and scale. The marginal posterior for μ is Student-t with 4 degrees of freedom, location 3 and scale $\sqrt{5/(2 \cdot 5)} = \sqrt{0.5}$.

Question 5: NIG posterior predictive distribution

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Do not stop after finding the posterior for the parameters. For a future observation, integrate the Normal sampling distribution over both the coefficient or mean uncertainty and variance uncertainty. Under the Normal–Inverse-Gamma model, the result is a Student t predictive distribution. Identify its location, scale and degrees of freedom and explain why it has heavier tails than a Normal plug-in prediction. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The data are $-1, 0, 1$, so $n = 3$, $\overline{y} = 0$, and $S = \sum_i (y_i - 0)^2 = 2$. The prior has $\mu_0 = 0$, $\kappa_0 = 1$, $a_0 = 2$, $b_0 = 1$. Therefore

$$\kappa_n = 1 + 3 = 4, \qquad \mu_n = \frac{1(0) + 3(0)}{4} = 0,$$

$$a_n = 2 + \frac{3}{2} = \frac{7}{2}, \qquad b_n = 1 + \frac{2}{2} + \frac{1(3)}{2(4)}(0 - 0)^2 = 2.$$

Thus $\sigma^2 \mid y \sim \text{InvGamma}(7/2, 2)$ and $\mu \mid \sigma^2, y \sim N(0, \sigma^2/4)$. The predictive degrees of freedom are $2a_n = 7$ and predictive scale is

$$s_* = \sqrt{\frac{b_n(\kappa_n + 1)}{a_n \kappa_n}} = \sqrt{\frac{2(5)}{(7/2)(4)}} = \sqrt{\frac{5}{7}} \approx 0.8452.$$

Hence $\widetilde{Y} \mid y \sim t_7(0, \sqrt{5/7})$. This is a predictive distribution for a new noisy observation, not merely the conditional posterior of the unknown mean.

11.4 Solutions: Type 4

Teacher's spoken explanation

Let us work through solutions: type 4 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the

parameterization used here is the same one used in the preceding derivation.

Question 3: MAP under a transformation

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. This problem is testing the Jacobian. Transform the posterior density into the new parameter, including the absolute derivative of the inverse transformation. Then maximize the transformed density. Do not simply transform the old MAP estimate: density maxima are not invariant under nonlinear reparameterization. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The Gamma$(2,1)$ density is $p_\theta(\theta) = \theta e^{-\theta}$ for $\theta > 0$. Its log density is $\log\theta - \theta$, with derivative $1/\theta - 1$, so the maximum occurs at $\widehat{\theta}_{\mathrm{MAP}} = 1$.

Set $\psi = \log\theta$, so $\theta = e^{\psi}$ and $\mathrm{d}\theta/\,\mathrm{d}\psi = e^{\psi}$. By change of variables,

$$p_\psi(\psi) = p_\theta(e^{\psi})e^{\psi} = e^{\psi}e^{-e^{\psi}}e^{\psi} = e^{2\psi - e^{\psi}}, \qquad -\infty < \psi < \infty.$$

The derivative of the log density is $2 - e^{\psi}$, which is zero at $e^{\psi} = 2$, hence $\widehat{\psi}_{\mathrm{MAP}} = \log 2$. This is not $\log(\widehat{\theta}_{\mathrm{MAP}}) = \log 1 = 0$. The Jacobian factor changes the density and therefore changes its mode. The result illustrates that a MAP estimate is parameterization-dependent.

Question 4: equal-tail Beta credible interval

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Identify the posterior Beta parameters, then define the lower and upper posterior quantiles. For a 95 percent equal-tail interval, use the 0.025 and 0.975 quantiles. Explain that these quantiles are generally obtained numerically or from a Beta quantile function; there is no requirement that the interval be symmetric around the posterior mean. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The posterior is Beta$(3,4)$. A central 90% equal-tail interval uses the posterior quantiles 0.05 and 0.95, not 0.025 and 0.975. The supplied quantiles are 0.1532 and 0.7287, so

$$C_{ET} = [0.1532, 0.7287].$$

By construction,

$$P(0.1532 \le \theta \le 0.7287 \mid y) = 0.90$$

(up to rounding of the endpoints). This is a posterior probability conditional on the observed data and the prior/model. It is not a statement about the long-run coverage frequency of a confidence-interval procedure.

Question 5: point estimate under asymmetric loss

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Write the loss function before selecting an estimator. Under asymmetric absolute loss, errors on one side cost more than errors on the other, so the posterior median is replaced by a posterior quantile determined by the relative loss weights. Differentiate the posterior expected loss or use the quantile characterization, then interpret why the estimate shifts toward the less costly side. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Let a be the action. If $a < \theta$, underestimation loss is $3(\theta - a)$; if $a > \theta$, overestimation loss is $(a - \theta)$. The posterior expected loss is

$$R(a) = 3\int_a^{\infty} (\theta - a)p(\theta \mid y)\,\mathrm{d}\theta + \int_{-\infty}^{a} (a - \theta)p(\theta \mid y)\,\mathrm{d}\theta.$$

Differentiating with respect to a gives

$$R'(a) = -3P(\theta > a \mid y) + P(\theta < a \mid y).$$

At an optimum, $R'(a) = 0$, so $F(a \mid y) = 3[1 - F(a \mid y)]$, which implies $4F(a \mid y) = 3$ and hence $F(a \mid y) = 0.75$. Therefore

$$\boxed{a^* = q_{0.75} = F^{-1}_{\mathrm{Beta}(4,2)}(0.75).}$$

Under this loss, underestimation is three times as costly as overestimation, so the optimal action is pushed upward to the 75th posterior percentile. It is not generally the mean or median; those correspond to different loss functions.

11.5 Solutions: Type 5

Teacher's spoken explanation

Let us work through solutions: type 5 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 3: Jeffreys' prior for a Poisson rate

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Compute the Fisher information for the Poisson rate from the score or the expected negative second derivative. Take its square root to obtain the Jeffreys prior kernel. State whether the resulting prior is proper on the positive real line. The posterior can still be proper after data, but do not confuse those two questions. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

For one observation $Y \sim \text{Poisson}(\lambda)$,

$$\ell(\lambda; y) = -\lambda + y \log \lambda - \log(y!), \quad \frac{\partial \ell}{\partial \lambda} = -1 + \frac{y}{\lambda}.$$

The score has mean zero and variance

$$\mathcal{I}(\lambda) = \text{Var}\left(-1 + \frac{Y}{\lambda}\right) = \frac{\text{Var}(Y)}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

Therefore Jeffreys' prior is

$$\pi_J(\lambda) \propto \sqrt{1/\lambda} = \lambda^{-1/2}, \quad \lambda > 0.$$

The integral $\int_0^\infty \lambda^{-1/2}\,\mathrm{d}\lambda$ diverges at infinity, so the prior is improper. For n independent unit-exposure observations with total count s, the likelihood kernel is $\lambda^s e^{-n\lambda}$. Multiplying by the prior gives $\lambda^{s-1/2}e^{-n\lambda}$, which is

$$\lambda \mid y \sim \text{Gamma}\left(s + \frac{1}{2}, n\right).$$

For $n > 0$, the posterior is proper because the Gamma shape and rate are positive.

Question 4: Jeffreys' prior for a Bernoulli probability

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. For Bernoulli data, derive Fisher information for theta and take its square root. The resulting density is proportional to theta to minus one half times one minus theta to minus one half, which is the Beta one-half, one-half prior. It favours values near the boundaries more than a uniform prior does, while remaining proper. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

For Bernoulli(p), the Fisher information is $\mathcal{I}(p) = 1/[p(1-p)]$. Thus

$$\pi_J(p) \propto [p(1-p)]^{-1/2},$$

which is Beta$(1/2, 1/2)$. With $z = 7$ successes and $n - z = 3$ failures,

$$p \mid y \sim \text{Beta}\left(7 + \frac{1}{2}, 3 + \frac{1}{2}\right) = \text{Beta}(7.5, 3.5).$$

The predictive probability of success on the next trial is the posterior mean:

$$P(\widetilde{Y} = 1 \mid y) = \frac{7.5}{7.5 + 3.5} = \frac{7.5}{11} \approx 0.6818.$$

Question 5: prior odds are not the Bayes factor

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Use the numbers to calculate prior odds and the Bayes factor separately. Then multiply them to obtain posterior odds. The key conceptual correction is that a Bayes factor measures the data's relative support for two models; prior odds express the starting preference. A Bayes factor cannot be read as the probability that a model is true. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The prior probability of H_1 is 0.10, so $P(H_0) = 0.90$. Prior odds in favor of H_1 are

$$\frac{P(H_1)}{P(H_0)} = \frac{0.10}{0.90} = \frac{1}{9}.$$

Multiplying by $\mathrm{BF}_{10} = 4$ gives posterior odds $4/9$. Therefore

$$P(H_1 \mid y) = \frac{4/9}{1 + 4/9} = \frac{4}{13} \approx 0.3077.$$

The value 0.4 is 4 times the prior probability 0.1, but Bayes' theorem updates odds, not probabilities by direct multiplication. The correct denominator must also account for the probability of H_0.

11.6 Solutions: Type 6

Teacher's spoken explanation

Let us work through solutions: type 6 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 3: diagnose a trace plot

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Describe what the plotted sequence does over iteration. Look for a stable horizontal band, frequent movement and no long-term trend. A drift, abrupt regime change or long flat segments suggests poor mixing or failure to reach stationarity. A single trace plot is not enough: mention multiple chains, density overlays and autocorrelation as complementary checks. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The three chains occupy different neighborhoods and do not move between them. Flatness within each chain only says that each chain has not visibly changed; it does not show that the chains have reached the same stationary distribution. Their starting values may have trapped them in different posterior regions or modes. The chains therefore provide evidence of poor mixing or non-convergence, not convergence.

At least four checks should be made:

1. **Trace plots:** after warm-up, all chains should fluctuate around a common stable range without persistent trends or long flat segments.
2. **Overlaid marginal density plots:** chain-specific histograms/densities should have similar centers, spreads, shapes and tails.
3. **Autocorrelation plots:** autocorrelation should decay rather than remain near one over many lags; a slow decay indicates poor mixing.
4. **Running means:** chain-specific running means should stabilize and agree within Monte Carlo uncertainty.

5. **Split-$\widehat{R}$ and ESS:** $\widehat{R}$ should be close to one and effective sample sizes should be adequate for the desired precision.

No single diagnostic proves convergence. Multiple dispersed chains and graphical summaries should be interpreted together.

Question 4: asymmetric Metropolis–Hastings proposal

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Use the complete acceptance ratio and retain the proposal-density correction. Write q(current given proposed) in the numerator and q(proposed given current) in the denominator. Only cancel these terms when the proposal is symmetric. This question checks whether the student understands the Hastings correction rather than memorizing the random-walk special case. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The current state is 0 and the proposal is 1. The target ratio is $\pi(1)/\pi(0) = 0.4/0.6 = 2/3$. The proposal ratio is $q(0 \mid 1)/q(1 \mid 0) = 0.2/0.8 = 1/4$. Thus

$$\alpha(0,1) = \min\left\{1, \frac{0.4(0.2)}{0.6(0.8)}\right\} = \frac{0.08}{0.48} = \frac{1}{6} \approx 0.1667.$$

If the proposal ratio were omitted, one would calculate $\min(1, 0.4/0.6) = 2/3$, which is incorrect. The asymmetric proposal visits state 1 much more often when leaving 0 than it proposes the reverse move, so the correction reduces acceptance to preserve the desired target distribution.

Question 5: effective sample size

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Start from the stored number of MCMC draws and the autocorrelation at successive lags. Use the integrated autocorrelation time to estimate effective sample size. Then calculate the Monte Carlo standard error of the posterior mean as the posterior standard deviation divided by the square root of ESS. Explain why ten thousand correlated draws can contain far less information than ten thousand independent draws. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Here $\rho_1 = 0.8$. The AR(1) approximation gives

$$\text{ESS} \approx 1000\frac{1 - 0.8}{1 + 0.8} = 1000\frac{0.2}{1.8} \approx 111.11.$$

With posterior standard deviation $s = 2$, the Monte Carlo standard error is approximately

$$\text{MCSE} \approx \frac{2}{\sqrt{111.11}} \approx 0.1897.$$

For 1000 independent draws, ESS = 1000 and the corresponding standard error is

$$\frac{2}{\sqrt{1000}} \approx 0.0632.$$

The correlated chain has roughly the information content of only 111 independent draws, so its Monte Carlo error is about three times larger. Running more iterations or improving mixing can increase effective sample size.

11.7 Solutions: Type 7

Teacher's spoken explanation

Let us work through solutions: type 7 carefully. First, identify the object we are trying to learn about and the probability model that connects it to the observations. Next, state the assumptions and the meaning of each symbol before manipulating the formula. As we develop this topic, do not treat the displayed expression as something to memorize without interpretation: explain what each term contributes, why the conditioning is appropriate, and what the final result tells us about uncertainty. At the end, check that the conclusion answers the question asked and that the parameterization used here is the same one used in the preceding derivation.

Question 3: derive the LASSO prior

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Begin with the Laplace density and take its negative logarithm. The result is a constant plus an absolute-value penalty. Combine this with the Gaussian regression negative log likelihood to recover the LASSO objective. Explain why a Laplace prior creates a cusp at zero and encourages sparse MAP estimates. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The Gaussian likelihood kernel is

$$p(y \mid \beta, \sigma^2) \propto \exp\left\{-\frac{\|y - X\beta\|^2}{2\sigma^2}\right\}.$$

The independent Laplace prior kernels multiply to

$$\pi(\beta \mid \lambda, \sigma^2) \propto \exp\left\{-\frac{\lambda}{\sigma^2}\sum_j |\beta_j|\right\} = \exp\left\{-\frac{\lambda}{\sigma^2}\|\beta\|_1\right\}.$$

The negative log posterior, ignoring constants, is therefore

$$\frac{1}{2\sigma^2}\|y - X\beta\|^2 + \frac{\lambda}{\sigma^2}\|\beta\|_1.$$

Multiplying by $2\sigma^2$ does not change the minimizer, so

$$\boxed{\widehat{\beta}_{\text{MAP}} = \arg\min_{\beta}\left\{\|y - X\beta\|^2 + 2\lambda\|\beta\|_1\right\}.}$$

The L_1 norm has a sharp corner at zero. For sufficiently strong penalty, the likelihood improvement from a small nonzero coefficient is not enough to offset the penalty, so the optimum can occur exactly at zero. This is the MAP/LASSO connection. The precise factor multiplying $\|\beta\|_1$ depends on the rate/scale used in the Laplace prior.

Question 4: spike-and-slab posterior inclusion

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. Write the prior inclusion probability and the two coefficient distributions: a point mass at zero for the spike and a continuous distribution for the slab. Calculate the marginal likelihood under inclusion and exclusion, then apply Bayes' theorem to the indicator. The final posterior inclusion probability is a probability about whether the variable is in the model, not merely whether its fitted coefficient is numerically large. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

The prior inclusion probability is $\pi = 0.20$, so prior exclusion probability is 0.80. The Bayes factor for inclusion over exclusion is

$$\mathrm{BF}_{10} = \frac{m_1(y)}{m_0(y)} = \frac{0.12}{0.03} = 4.$$

Prior inclusion odds are $0.20/0.80 = 1/4$. Posterior inclusion odds are $4(1/4) = 1$, hence

$$P(\gamma = 1 \mid y) = \frac{1}{1+1} = 0.50.$$

Equivalently,

$$P(\gamma = 1 \mid y) = \frac{0.20(0.12)}{0.20(0.12) + 0.80(0.03)} = \frac{0.024}{0.024 + 0.024} = 0.50.$$

The spike-and-slab prior encodes exclusion by assigning a point mass at exactly $\beta = 0$ when $\gamma = 0$, and a continuous slab distribution when $\gamma = 1$. The posterior inclusion probability is the probability of the indicator being one after seeing the data.

Question 5: empirical Bayes and profile likelihood

Teacher's spoken explanation

Teacher: Let us solve this one as a classroom demonstration. First integrate out the group-level effects to obtain the marginal likelihood for the hyperparameters. Empirical Bayes estimates those hyperparameters by maximizing this marginal likelihood. For profile likelihood, instead fix the parameter of interest and maximize over nuisance parameters. Emphasize the distinction: marginal likelihood integrates; profile likelihood maximizes. They are not interchangeable operations. Now write the derivation on the board in a clean sequence, pause after each line, and ask what assumption justifies that step. Finish by checking the units, parameter restrictions and interpretation of the answer.

Let

$$v_j(\tau^2) = \tau^2 + \frac{\sigma^2}{n_j}.$$

The marginal likelihood is the product of independent Normal densities:

$$m(\overline{y}_1, \ldots, \overline{y}_J \mid \mu, \tau^2) = \prod_{j=1}^{J} \frac{1}{\sqrt{2\pi v_j(\tau^2)}} \exp\left\{ -\frac{(\overline{y}_j - \mu)^2}{2v_j(\tau^2)} \right\}.$$

For a fixed τ^2, the log marginal likelihood depending on μ is

$$\ell(\mu; \tau^2) = -\frac{1}{2} \sum_{j=1}^{J} \frac{(\overline{y}_j - \mu)^2}{v_j(\tau^2)} + C(\tau^2).$$

Differentiate and set equal to zero:

$$\frac{\partial \ell}{\partial \mu} = \sum_{j=1}^{J} \frac{\overline{y}_j - \mu}{v_j(\tau^2)} = 0.$$

Therefore

$$\boxed{\widehat{\mu}(\tau^2) = \frac{\sum_j \overline{y}_j / v_j(\tau^2)}{\sum_j 1/v_j(\tau^2)}.}$$

The profile likelihood for τ^2 is obtained by substitution:

$$\boxed{L_p(\tau^2) = m(\overline{y}_1, \ldots, \overline{y}_J \mid \widehat{\mu}(\tau^2), \tau^2).}$$

An empirical Bayes analysis maximizes this marginal likelihood over τ^2 (and profiles or estimates μ), then plugs the estimated hyperparameters into the group-level posterior. A fully Bayesian analysis instead puts a hyperprior on (μ, τ^2) and averages the group posterior over the hyperparameter posterior. The former is a plug-in method; the latter propagates hyperparameter uncertainty.

Teacher's spoken explanation

That completes our journey. We began with one unknown probability and a small amount of evidence. We learned how prior beliefs and likelihood combine, how conjugate families make updating transparent, how a posterior can be summarized and used for prediction, how hypotheses can be compared, and how priors affect inference. We then faced posteriors too complicated for hand calculation and learned to use MCMC carefully. Finally, we built regression models and allowed groups to share information through a hierarchy. The mathematics has become more elaborate, but the governing idea has not changed: specify a probability model, condition on the data, and reason transparently about what remains uncertain. That is the unifying principle of Bayesian inference.

Teacher's spoken explanation

Before you leave this course, I want you to remember the path rather than a disconnected catalogue of equations. We began by asking how evidence changes uncertainty. We learned to express that change through Bayes' theorem. We found conjugate models that let us update exactly, then learned how to summarize the posterior without confusing an estimate with the full distribution. We used marginal likelihoods to compare hypotheses, examined the role of priors, and developed simulation methods for posteriors that cannot be handled neatly by hand. We extended the model to several predictors and finally to groups with shared structure.
When you face an unfamiliar question in the examination, do not panic because the story or notation looks new. Translate it. What is observed? What is unknown? What is the sampling model? What prior or hyperprior has been specified? Is the task to update, estimate, predict, compare, simulate, select or pool? Once you answer those questions, you have found your place in the course. Then write the mathematics carefully, state the assumptions, and interpret the result in words. That is how a statistician thinks, and that is the standard I want you to carry into the examination.

Final self-test before the examination

Before the examination, confirm that you can do each of the following from a blank page:

1. State Bayes' theorem and identify prior, likelihood, evidence and posterior in a word problem.
2. Derive the Beta, Gamma and Dirichlet conjugate updates by multiplying kernels.
3. Complete the square to derive the Normal–Normal and Normal–Inverse-Gamma updates.
4. Distinguish posterior mean, median, mode/MAP and MLE, and select a Bayes estimator from a loss function.
5. Construct an equal-tail credible interval from posterior quantiles and explain its meaning.
6. Compute posterior odds from prior odds and a Bayes factor, including for a composite hypothesis.
7. Derive Jeffreys' prior from Fisher information and check posterior propriety under an improper prior.
8. Write a Gibbs update and calculate a Metropolis–Hastings acceptance probability for both symmetric and asymmetric proposals.
9. Read trace, density, autocorrelation and running-mean plots and explain ESS.
10. Derive the ridge and LASSO MAP objectives from their respective priors, and explain spike-and-slab inclusion probabilities.
11. Explain partial pooling, empirical Bayes, hierarchical Bayes and profile likelihood without conflating them.

A final exam-writing habit

At every step, name the distribution and its parameters. "The posterior is Gamma" is incomplete; write "$\lambda \mid y \sim \text{Gamma}(a + \sum y_i, b + n)$ under the shape–rate convention." "The credible interval is 95%" is incomplete; state the endpoints/quantiles and whether it is equal-tail or HPD. This small habit prevents avoidable loss of marks.